Applying Automotive EDR Data to Traffic Crash Reconstruction

Introduction

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in (Ctrl+Plus)

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Learning Objectives (brochure)

- Describe EDR sensor operation, recording interval, accuracy & duration, and time latency and articulate the limitations of applying the data to crash analysis
- Calculate Min and max speeds prior to loss of control or braking, and at impact, based on the last accurate EDR pre-crash speed data point.
- Evaluate EDR vs. actual ground speed for specific vehicle operational conditions and vehicle equipment modifications.

Learning Objectives (brochure)

- Calculate speed at impact and closing speeds by combining EDR Delta V data with normally collected scene evidence and vehicle data such as post crash travel distance, departure angle, drag factor, and vehicle weights. Apply data to inline rear end, head on, and angular collisions.
- Reconcile EDR data with other physical evidence and combine to narrow speed ranges
- Use time-distance to overlay EDR data on scene maps/diagrams to show where critical driving inputs were made versus inputs required to avoid collisions.

More Learning Objectives

- Authoritatively determine if a recording is from the crash we are investigating or not.
- Understand what special circumstances may affect Delta V accuracy and interpretation
- Determine if cause of crash is speed, inattention, or a combination of both.
- How to present data to win EDR Frye and Daubert hearings

SAE Learning Assessment

- You don’t have to take it, but you have to take it if you want a certificate of completion to get CEU’s
- It’s online, they send you the link to it after the class. It’s open book, open notes, but take it SOON before you forget details.
- 20 questions multiple choice – but don’t compare answers with anyone. Everyone gets a different test – question order and answer order is randomized

Introduction -Main Messages

1. EDR's improve recon **accuracy** & get to **causation** – use with **all other** available scene evidence – but if other evidence is lacking, EDR can be used a PRIMARY EVIDENCE.
2. The EDR strength is to give us speed **PRIOR** TO *BRAKING* OR *LOSS OF CONTROL, &* DRIVER BEHAVIOR on gas and brake pedal. Modern ABS brakes TOOK AWAY your tire marks, only EDR can "give you back" speed prior to braking. Newer data elements give us EVEN MORE INSIGHT into precrash behavior.
3. Event Data Recording capability and data accessibility ***varies widely*** by **manufacturer, model *and* model year.** Coverage keeps growing with MORE AND BETTER DATA available with each new model released.

Main Messages

4. **Minimum** Data in **new** vehicles equipped with EDR is ***NOW STANDARDIZED***. NHTSA EDR Part 563 rule took effect Sept 2012, ensures public access to data, and that data survives in severe crashes.
5. Proposed rule to REQUIRE EDR on 100% of passenger vehicles by 9/2014 was *NOT implemented!* *Coming ..”soon”!*
6. EDR’s will NEVER replace human beings in figuring out what really happened.
7. PRIVACY – New US law Dec 2015 requiring owner permission or a warrant.

Keeping EDR's in Perspective

- Of 289 million vehicles registered in the US & CA, **50%** have an EDR with data accessible by the Bosch CDR system
- 88% of 2016's are supported by Bosch.
- Another 11% have EDR accessible by other tool. **99%** have EDR.
- In two car collisions, on average one will have a supported EDR. With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles.

Scope of EDR Investigators

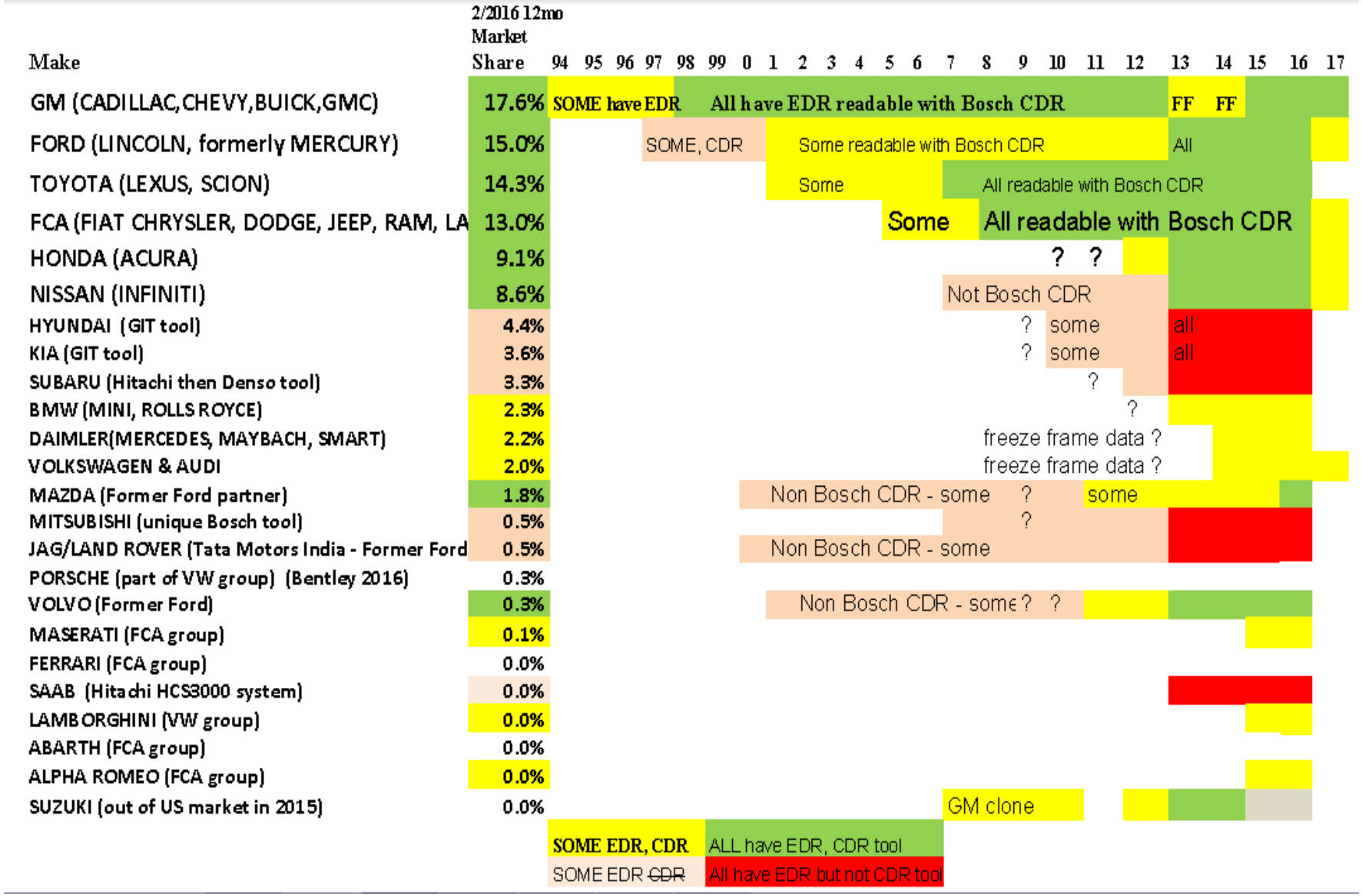
- 1800 Bosch CDR Active Software subscriptions.
- Half Law Enforcement
- Half private, mostly reconstructionists
- Insurance use still very small so far
- One survey put the average user as doing 4 images per month
- Europe and Australia – but only a small # of vehicles covered there. New Korean reg Dec 2015. Japan, China next.

US EDR Presence & Tool Used

Manufacturer	12 months Sales end 2/16	Market Share	2016 EDR?	Tool?
GM	3,079,772	17.6%	YES	Bosch CDR
Ford	2,634,491	15.0%	YES	Bosch CDR
Toyota	2,498,889	14.3%	YES	Bosch CDR
Chrysler Group	2,273,230	13.0%	YES	Bosch CDR
Honda	1,598,383	9.1%	YES	Bosch CDR
Nissan	1,499,020	8.6%	YES	Bosch CDR
Hyundai	762,720	4.4%	YES	GIT tool co.
Kia	631,531	3.6%	YES	GIT tool co.
Subaru	583,617	3.3%	YES	Denso 2016+, was Hitachi
BMW	400,064	2.3%	2015	Bosch CDR
Mercedes-Benz	380,690	2.2%	2014	Bosch CDR
Volkswagen	342,626	2.0%	2015	Bosch CDR
Mazda	314,510	1.8%	YES	Bosch CDR
Audi	202,774	1.2%	2015	Bosch CDR
Mitsubishi	95,450	0.5%	YES	OTC/Bosch SPX
Tata (Jag, LR)	87,120	0.5%	YES	OTC/Bosch SPX
Geely (Volvo)	71,744	0.4%	YES	Bosch CDR
Porsche	52,532	0.3%	NO	N/A
Industry Total	17,509,163	100.0%	99.3%	Bosch CDR 87.0%
updated April 4 2016				Non CDR 12.3%

EDR Phase-in by Mfr

Data is incomplete for smaller mfrs with no public tool



2016 Recording Capability*

*based on 2012

	Part 563 Table 1					Part 563 Table 2						Beyond 563								
Mfr	Event Complete	Key Cycles	Long. ΔV	Speed/Brake/ Throttle or Accel	Multi Event	Lateral ΔV	RPM	Steering Angle	ACM Acceleration	ABS on/off	Roll Angle	ESC Long Accel	ESC Lat Accel	Yaw Rate	Tire Pressure	Panic Brake Assist	Wheel Speeds	Roll Rate	Brake Pressure	Day/Date/Time of Crash
GM	X	X	X	5@0.5 TA	X	X	X	-	X, Y, Z 300@2	X		-			LO	-	-	1 @ 10ms		
Ford	X	X	X	5@0.5 A	X	X	X	5@0.1	-	X		5@0.1		5@0.1	LO		-	5@0.1		
Chrysler	X	X	X	5@0.1 TA	X	X	X	5@0.1	-	X		-		5@0.1	PSI	X	X			
Toyota	X	X	X	5@0.5 TA	X	Side	X	5@0.5	-	?	?	5		5@0.5	-	-	-	In Roll	5@0.5	
Honda	X	X	X	5@0.5 TA	X	X	X	5@0.5	X, Y, Z 250@10	X	-1+5@0.1	-		-	-	-	-			
Nissan	X	X	X	5@0.5 A	X	X	X	5@0.5	X, Y 250@10	-	-	-		-	-	-	?			
Mazda	X	X	X	5@0.5 T	X	X	X	5@0.5	X, Y 250@10	-							-			
Volvo	X	X	X	5@0.5T	X	-	-	-	X,Y,Z 250@10	-	-1+0.3@0.1									
BMW	X	X	X	5@0.5A	X		X	5@0.5	X, Y 300@10ms	X	-									
Mercedes	X	X	X	5@0.5A	X															
VW	X	X	X	5@0.5A	X	X														
Hyundai/Kia	X	X	X	5@0.5	X	X	X	5@0.5	X,Y,Z 250@10	X	-1+5@0.1	-		-	-	-	-			
Subaru	X	X	X	5@0.5A	X	X	X	5@0.5		X										
Mitsubishi	X	X	X	5@0.5A	X	X	X		X, Y 250@10											

EMERGING TRENDS

- In 17.0, Bosch identifies the license holder who's software downloaded the initial .cdrx file, and the license holder who interpreted the data of the current PDF.
- As CDR goes worldwide, this is necessary to reduce the risk of piracy in developing nations (or entering markets with reputations for abuse of intellectual property licensing).

EMERGING TRENDS

- More manufacturers are becoming concerned about cyber security, of hackers trying to take over vehicle control systems.
- Many are installing “gate-way” modules or features to protect ECU’s from hacking.
- For CDR DLC users, “back powering” may not work on newer vehicles by powering the ACM fuse alone. Direct to module may become required on some.

Overview by Manufacturer

Honda/Acura CDR 17.1 Coverage 23%

Model	Location	2011	2012	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR cable number								
ACURA								
RL	Under dashboard center		547					
TL	Under dashboard center		547					
TSX	Under dashboard center		547					
ZDX	Under dashboard center		547					
MDX	Under dashboard center		547		387/789			387/810
NSX	Under dashboard center							387/789
RDX	Under dashboard center			547			387/789	
ILX	Under dashboard center			547			387/789	
RLX	Under dashboard center				387/789			
TLX	Under dashboard center					387/789		
HONDA								
ACCORD	Under dashbord center			387/789			387/810	
CIVIC 2 DR	Under dashbord center		547				387/810	
CIVIC 4 DR	Under dashbord center		547	387/789			387/810	
CROSSTOUR	Under dashbord center		547					
CR-V	Under dashbord center		547				387/789	387/810
CR-Z	Under dashbord center		547					
FCX Clarity	Under dashbord center							387/810
FIT	Under dashbord center	EV	547			387/789		
HR-V							387/789	
INSIGHT	Under dashbord center		547					
ODYSSEY	Under dashbord center		547		387/789			387/810
PILOT	Under dashbord center		547				387/789	387/810
RIDGELINE	Under dashbord center		547					387/810

NO time series EDR But may have Delta V single value

Honda Precrash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

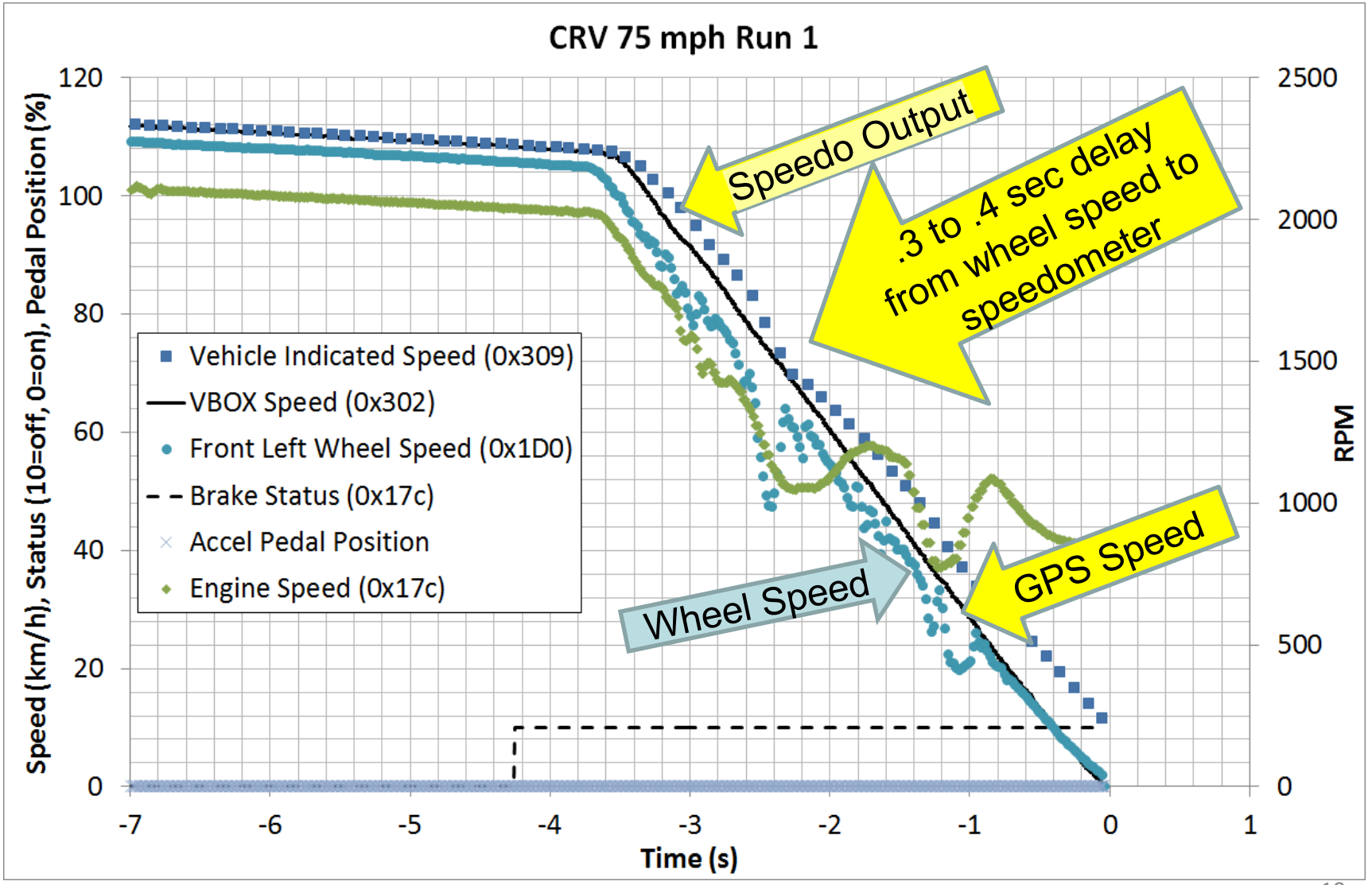
Same unless brake or stability control override

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	PCM Derived Accelerator Pedal Position, % full	Service Brake (On, Off)	ABS Activity (On, Off)	Stability Control (On, Off, Engaged)	Steering Input (deg)	Engine RPM	Accelerator Pedal Position, % full
-5.0	105 [169]	93	Off	Off	On Non-Engaged	RTFDL Pos=Left	5,800	93
-4.5	106 [170]	93	Off	Off	On Non-Engaged		5,800	93
-4.0	106 [170]	93	Off	Off	On Non-Engaged		5,900	93
-3.5	107 [172]	93	Off	Off	On Non-Engaged	0	5,900	93
-3.0	108 [174]	26	Off	Off	On Non-Engaged	0	5,300	26
-2.5	108 [174]	85	Off	Off	On Non-Engaged	0	4,600	85
-2.0	108 [174]	0	On	Off	On Non-Engaged	-20	4,600	0
-1.5	108 [174]	0	On	Off	On Non-Engaged	-5	4,400	0
-1.0	103 [166]	0	On	On	On Non-Engaged	-20	3,900	0
-0.5	103 [166]	0	On	On	On Non-Engaged	0	3,100	0
0.0	90 [145]	0	On	On	On Non-Engaged	19	2,400	0

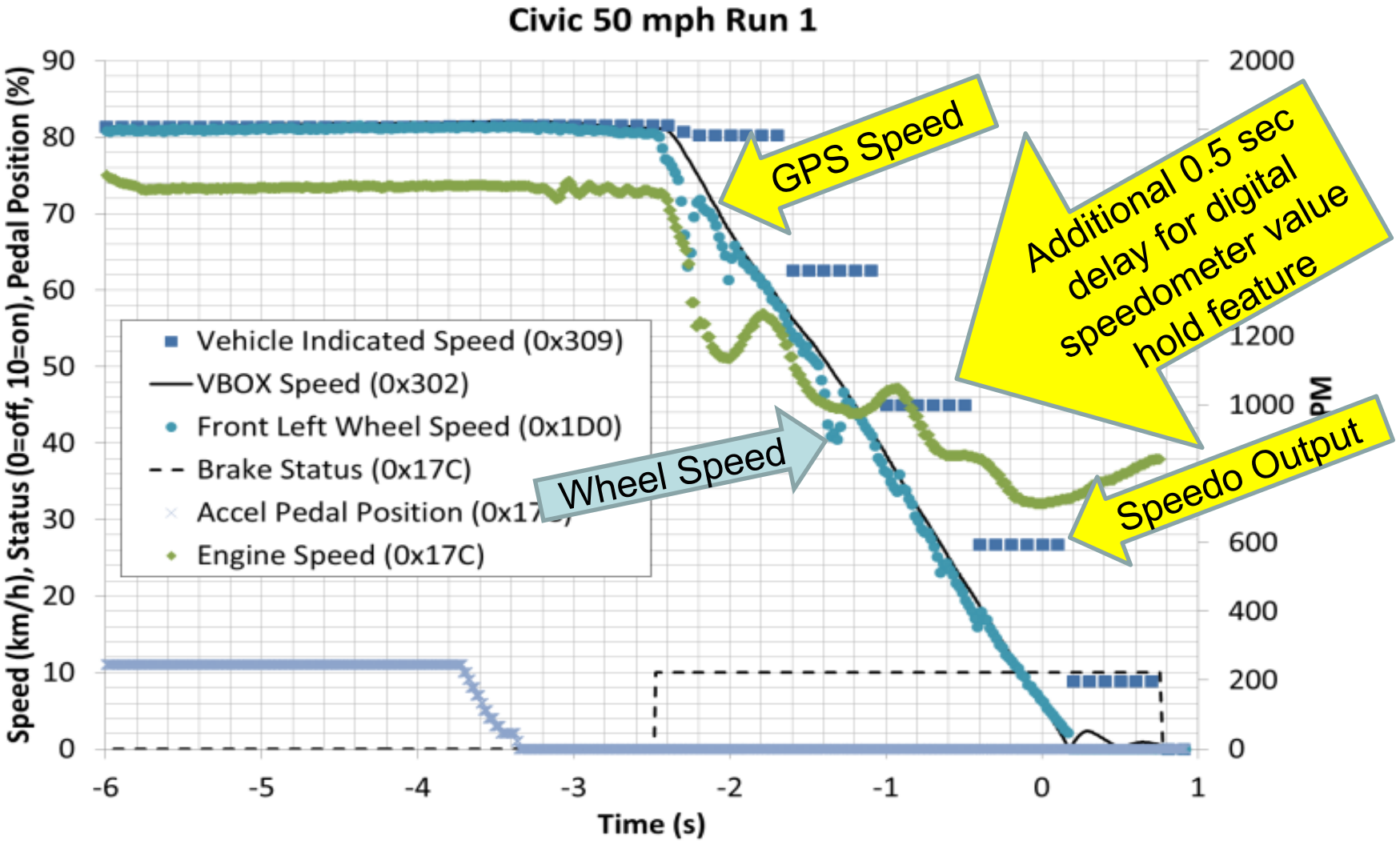
UNIQUE TO HONDA

- Most manufactures record the speed calculated in the powertrain control module as it is being sent TO the speedometer on the CAN bus.
- Honda records the speed being displayed ON the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
- The last data point before impact may be delayed more than it is from others

2012 CR-V Speed Reporting Delay



2012 Civic Speed Reporting Delay



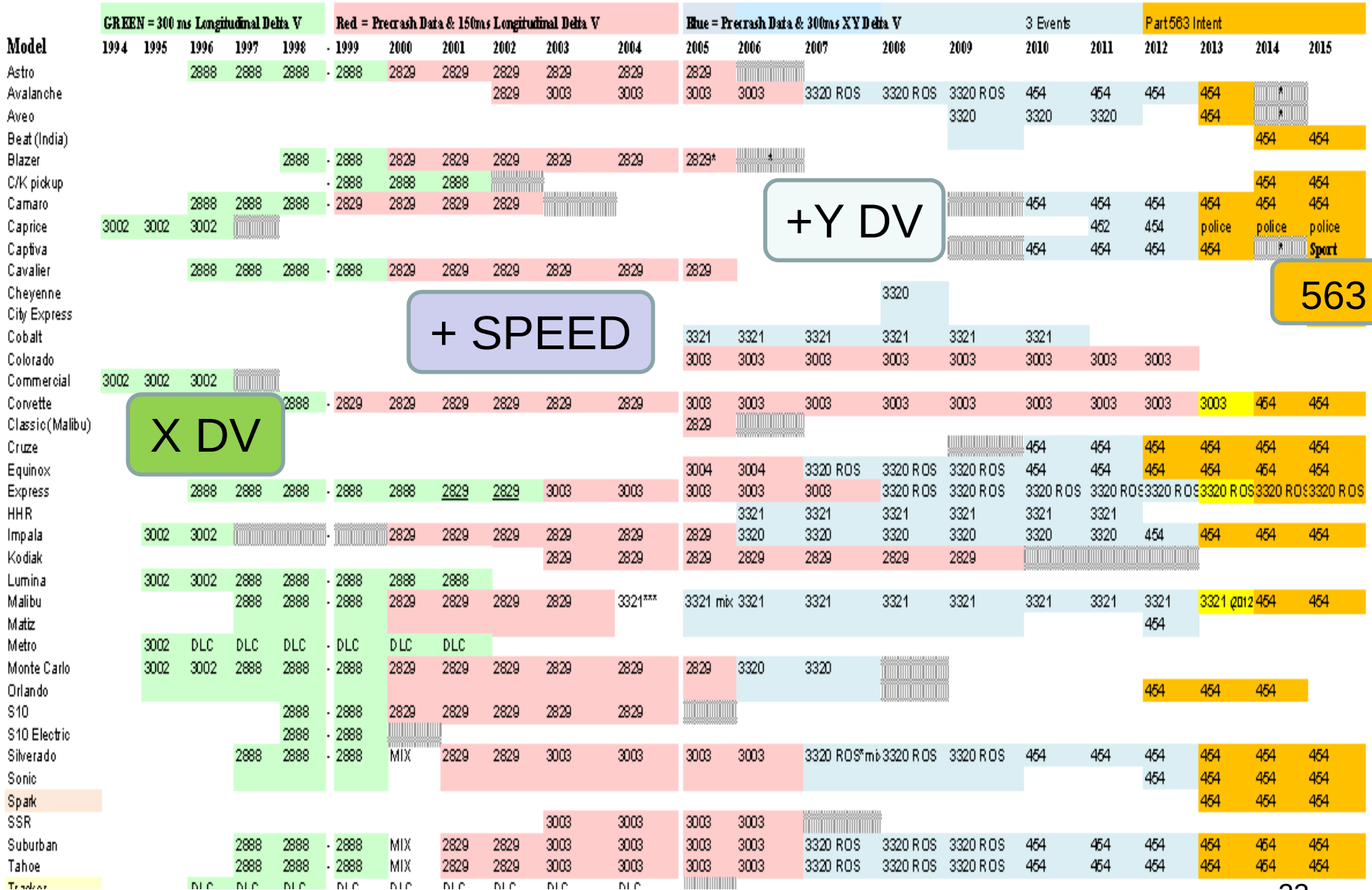
Honda Speed at Impact Calcs

- For ANALOG speedometer, assume last data point is SAMPLED 0.1 sec before AE, plus 0.3 sec filtering delay, adds up to 0.4 sec max delay from last data point to crash, use that in “MIN” column braking calcs.
- For DIGITAL speedometer, assume data is SAMPLED 0.1 sec before AE, plus 0.3 sec filtering delay, plus added 0.5 sec hold time. Add these to get 0.9 sec and use in “min” column braking calcs.

GM

- GM has so many models and brands, and for so many model years since 1994, the full timeline chart for just Chevrolet following is hard to read.
- A second chart follows showing only Chevrolet 2007 and later to make it more readable – the full GM chart is on my website www.ruthconsulting.com (as are the other manufacturer charts).

Chevrolet SDM Phase In **83%**



Chevrolet Recent Changes **83%** 563

1 animation

	3 Events			Part 563 Intent					
Model	2009	2010	2011	2012	2013	2014	2015	2016	2017
Astro									
Avalanche	3320 ROS	454	454	454	454	*			
Aveo	3320	3320	3320		454	*			
Beat (India)						454	454		
Blazer									
C/K pickup									
Camaro		454	454	454	454	454	454	454	454
Caprice			452	454	police	police	police	police	police
Captiva		454	454	454	454	*	Sport	Sport	
Cavalier									
Cheyenne									
City Express							780	780	780
Cobalt	3321	3321							
Colorado	3003	3003	3003	3003			454	454	454
Commercial									
Corvette	3003	3003	3003	3003	3003	454	454	454	454
Classic (Malibu)									
Cruze		454	454	454	454	454	454	454	454
Equinox	3320 ROS	454	454	454	454	454	454	454	454
Express	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454
HHR	3321	3321	3321						
Impala	3320	3320	3320	454	454	454	454	454	454
Kodiak	2829								
Lumina									
Malibu	3321	3321	3321	3321	3321 (2012	454	454	454	454
Matiz				454					

older family

Freeze Frame

GM Red Group Precrash Data
Blue Group similar (some 5@0.5)

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle
-5	96	2816	84
-4	97	3712	100
-3	93	3776	100
-2	56	1920	0
-1	13	640	0

Seconds Before AE	Brake Switch Circuit Status
-8	OFF
-7	OFF
-6	OFF
-5	OFF
-4	OFF
-3	OFF
-2	ON
-1	ON

GM 563 PRE-CRASH DATA

Pre-Crash Data -5.0 to -0.5 sec (Event Record 1)

Times (sec)	Accelerator Pedal, % Full (Accelerator Pedal Position)	Service Brake (Brake Switch Circuit State)	Engine RPM (Engine Speed)	Engine Throttle, % Full (Throttle Position)	Speed, Vehicle Indicated (Vehicle Speed) (MPH [km/h])
-5.0	0	On	640	2	0 [0]
-4.5	0	On	640	3	0 [0]
-4.0	0	Off	640	3	0 [0]
-3.5	21	Off	704	16	1 [1]
-3.0	21	Off	1408	20	1 [2]
-2.5	23	Off	1600	23	3 [5]
-2.0	25	Off	1856	25	6 [9]
-1.5	25	Off	2048	25	8 [13]
-1.0	0	Off	2048	21	10 [16]
-0.5	0	Off	1856	5	11 [17]

GM Recent Changes

- Some 2010+ SDM's hold up to 3 events and do not require replacement after a D. Events numbered vs. labeled D/ND. Events may not be in order
- Acceleration data (see clipping!)
- Black plastic SDM housings
- Freeze frame data in 2013 Corvette, no precrash in 2013 Express/Savanna after 9/1/2012

Other New GM News

- Some product liability data elements not relevant to reconstruction ***moved*** from CDR main report into **E**ngineering **T**ranslation **R**eports. Other elements do not display if not equipped (used to say “if equipped”)
- Euro Pedestrian protection – if you have it in your US car, you’ll get data (not many do yet).
- Known “Features” noted in Data Lims

563 EDR features - Diagnostic Info

BOSCH

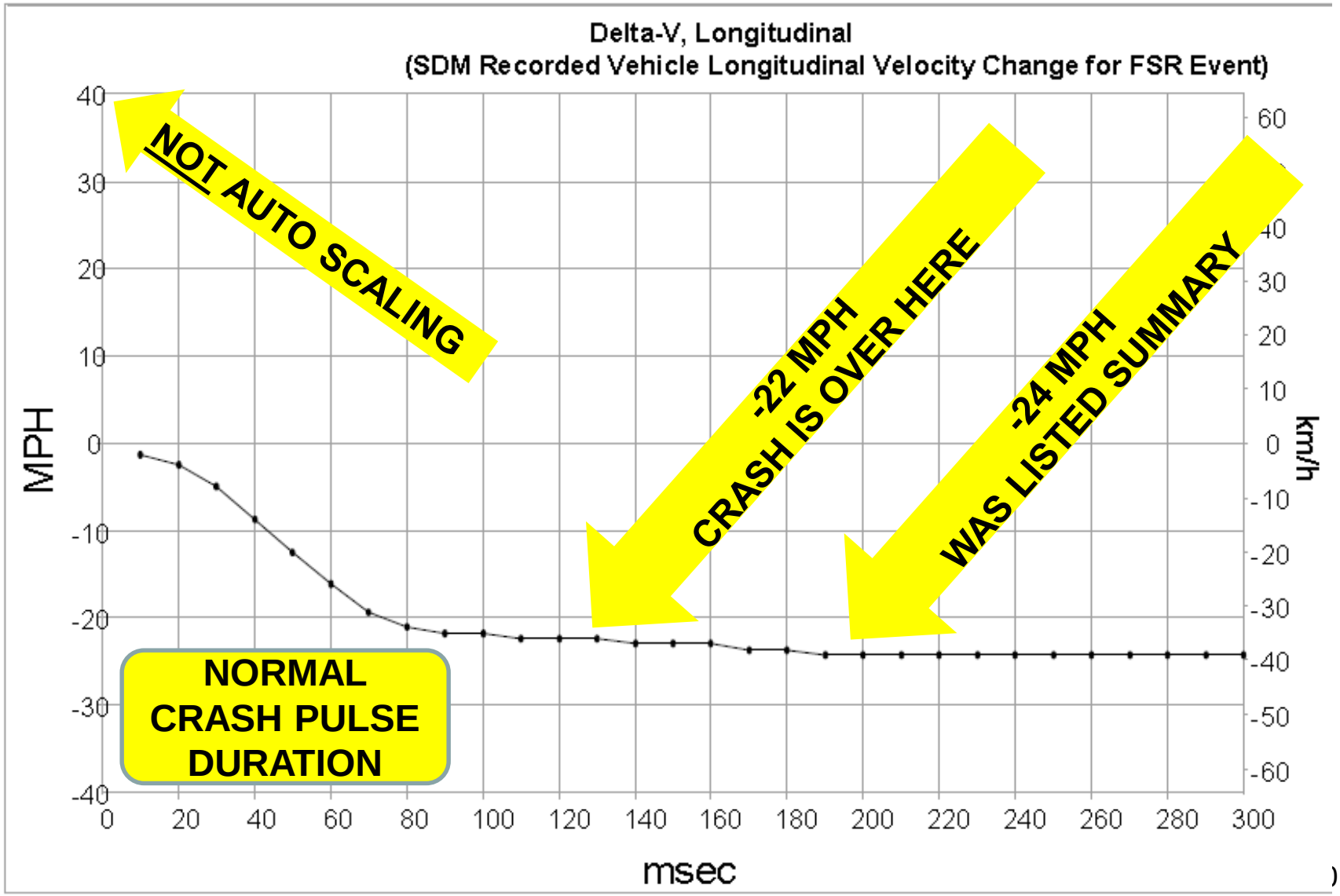
CDR CRASH DATA RETRIEVAL

DTCs Present at Time of Event (Event Record 1)

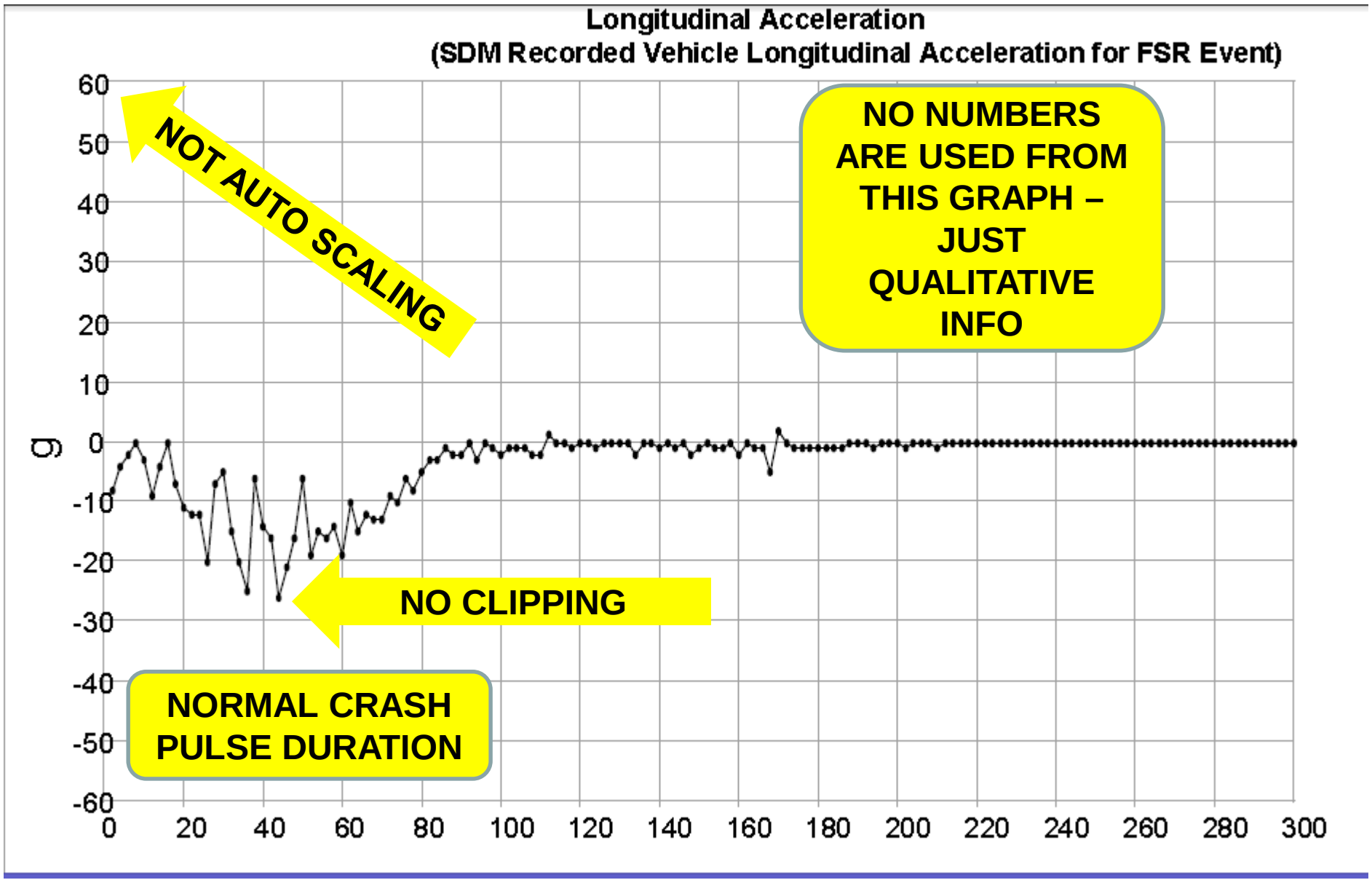
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- This is present on all new GM's if airbags deployed – it does not mean anything was wrong with the system before deployment

563 Longitudinal ***DV*** graph



563 Longitudinal ***Accel*** Graph



GM Ignition Switch Recall

- The GM Ignition Switch Recall is on 05-10 Cobalt, 06-11 HHR, 07-10 G5, 06-10 Solstice, 07-10 Sky, and 03-07 Ion. The key ring may pull on the ignition key causing it to rotate out of the run position, resulting in a loss of ignition power.
- In rare cases, collisions have followed the loss of ignition power, and some allege the recall condition caused the crash.
- EDR events, or lack of events, may give clues if the recall condition was present prior to an event.

GM Ignition Switch Recall

- The recall is not tied to any particular ACM family, but many of the models have the SDM “Epsilon” module, listed in the “Blue Group” on the chartrs. Others use the older SDM “DW” “Red Group”.
- The older SDM-DW will not make a recording if power is lost.
- The Epsilon has a capacitor that keeps it alive and may still record events for 3.25 to 4.50 seconds after key power is lost. lost.

How the GM vehicles work when ignition power is lost

- If the airbag deployment algorithm has enabled before ignition power is lost, the airbags can still deploy.
- Within 150ms after ignition power loss, the airbag deployment algorithms are disabled and any events that occur will be classified as NON deployments.

Possible Indications of Ignition Power Loss prior to impact in EDR

- “Vehicle Power Mode Status” -“Accessory” vs. “Run”
- “Run/Crank Ignition Switch Logic Level” shows “Inactive” vs. “Active”
- “Automatic Passenger SIR Suppression System Validity Status at AE” shows “Invalid” instead of “Valid”.
- Vehicle speed, RPM, throttle and other pre-crash data points close to impact show “0” or “invalid”.
- No event recorded if the power loss was more than 3.25-4.5 seconds before impact.
- Incomplete recordings if capacitor runs out

Check the ETR if data is invalid

- Any one of a time series being invalid can cause the CDR main report to show all the data points in that series as “invalid”.
- Open your .cdrx file. Under the file menu save it as an “Engineering Translation Report” or “ETR”. (Blue group and later)
- Review the ETR, it may show translations for precrash data listed as “invalid” in the main report
- If power was lost, precrash data will likely display as default zero values

For More Information

- Refer to the article “CDR Report data from vehicles subject to the GM Ignition Switch Recall with the “Epsilon” ACM” by W.R. “Rusty” Haight in the fall 2014 issue of Collision Magazine.
- Haight conducted staged collisions where the ignition power was cut before impact simulating the recall condition and presents screen shots showing the EDR data gathered.
- A new paper from GM is coming out soon on this issue

Ford CDR 17.1 Coverage 54%

	Model Year	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
FORD/MERCURY	500/Montego/Freestyle									~~CDR~~/PCM	-	-										
	Cmax																	783				
	Continental	NO		~~CDR~~	-	RCM																
	Cougar	NO		~~CDR~~	-	-	-															
	Crown Vic/Grand Marquis	NO	~~CDR~~	-	-	RCM				PCM/RCM												
	Econoline	~~CDR~~	-	-	-	-	-	-	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	384								
	Edge											PCM/RCM				384					783*	
	Escape/Mariner					RCM				~~CDR~~	-	-	-	384			384	783				
	Escort/ZX2	NO	~~CDR~~	-	-	RCM																
	Excursion				~~CDR~~	RCM																
	Expedition	NO			~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	-	-	384				783		
	Explorer Sport (2 dr)				00.5 ->	RCM		~~CDR~~														
	Explorer/Mountaineer	NO					~~CDR~~	-	~~CDR~~/PCM	-	PCM/RCM			~~CDR~~/PCM	-	384		783				
	F150	NO		~~CDR~~	-	RCM			~~CDR~~/PCM	-	-	-	-	384						783*	?	?
	Fiesta															384			783			
	Flex													~~CDR~~	384							
	Focus				~~CDR~~	-	-	-	-	-	-	-	RCM/~~PCM~~				597			783		
	Fusion/Milan										PCM/RCM		~~CDR~~	-	384			783*				823
	Mustang	NO	NO	~~CDR~~	~~CDR~~	RCM			RCM	~~CDR~~/PCM			~~CDR/PCM~~	-	384					783*		
	Ranger	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM						~~CDR~~	-	-	384							
	Sport Trac (pickup bed)				00.5 ->	RCM		~~CDR~~	-	-	-	PCM/RCM		PCM/~~RCM~~		~~CDR~~						
	Super Duty F250+		~~CDR~~	~~CDR~~	~~CDR~~	RCM				RCM/PCM**		PCM**	~~CDR/PCM~~	-	-	384						783
	Taurus/Sable	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM	~~CDR~~	-	-	-	-		~~CDR/PCM~~	-	384			783				
	T-bird	NO	NO			RCM		RCM/PCM														
	Transit																					783
	Transit Connect														~~CDR~~	-	-	778*	783			
	Windstar/Freestar/Monterey	NO	NO	~~CDR~~	-	RCM			~~CDR~~	-	-	-										
LINCOLN	Aviator (Explorer)							~~CDR~~	-	-		~~CDR~~/PCM										
	Continental			~~CDR~~	-	RCM																823
	LS				~~CDR~~	RCM		~~CDR~~/PCM	-	-	-											
	Mark LT (F150 platform)										~~CDR~~/PCM	~~CDR/PCM~~	-									
	MKC (Escape Platform)																			783		
	MKS (Taurus Platform)														384			783				
	MKZ, Zephyr (Fusion platform)										PCM/RCM		~~CDR/PCM~~	-	384			783	783*			823
	MKT '10 (Flex Platform)													~~CDR/PCM~~	384	384						
	MKX (Edge Platform)											PCM/RCM				384					783*	
	Town Car (Cr Vic)	NO	~~CDR~~	-	-	RCM				PCM/RCM												
	Navigator (Expedition)		NO		~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	~~CDR/PCM~~	-	-	-	384				783		

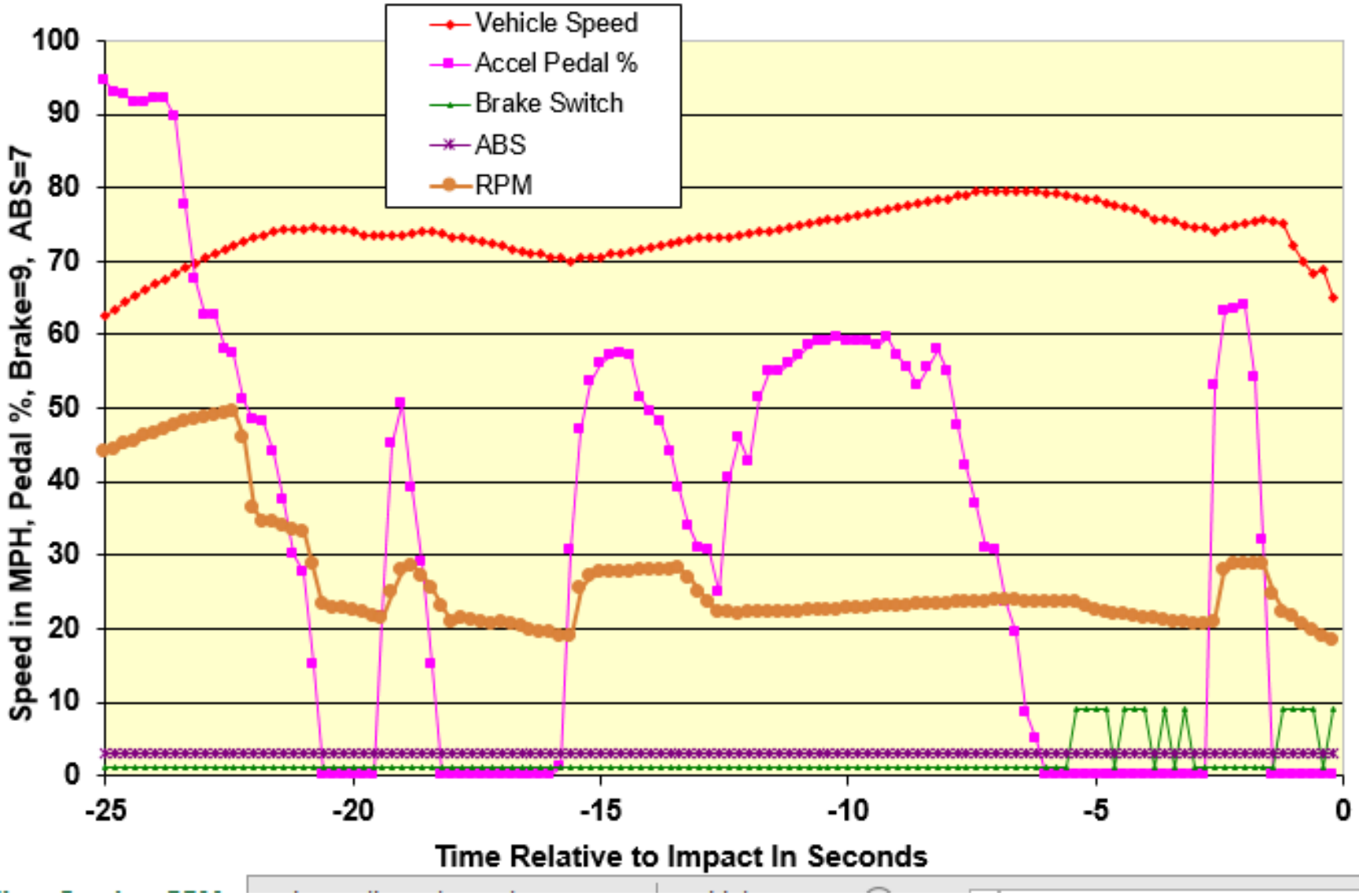
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SPEED
In ND's

X DV

+SPEED
In deployments

Typical Ford PCM Data Deploy



06 Ford ACM Precrash

Pre-Crash Data (First Record)

Time (sec)	-5	-4	-3	-2	-1
Accelerator Pedal (%)	7	7	7	7	1
Vehicle speed (MPH [km/h])	49.2 [79.2]	48.9 [78.8]	48.6 [78.3]	48.4 [77.9]	48.1 [77.4]
ABS event in progress	No	No	No	No	No
TCS engine event in progress	No	No	No	No	No
Brake depressed	No	No	No	No	No
OCS classification	Empty	Empty	Empty	Empty	Empty

09+ Ford **Pre- Crash Speed** Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (First Record)

Times (sec)	Speed vehicle indicated	Accelerator pedal, % full	Service brake, on/off	Engine rpm
- 5.0	36.0 [58.0]	0	OFF	0
- 4.5	37.3 [60.0]	0	OFF	0
- 4.0	37.9 [61.0]	0	OFF	0
- 3.5	38.5 [62.0]	0	OFF	0
- 3.0		0	OFF	0
- 2.5		0	OFF	0
- 2.0	39.2 [63.0]		OFF	0
- 1.5	39.2 [63.0]		OFF	0
- 1.0	39.2 [63.0]		OFF	
- 0.5	39.2 [63.0]	0	OFF	
0.0	39.2 [63.0]	0	OFF	

MPH (KPH)

NOT Throttle!

Exports to CSV!

09+Ford Stability Control Data

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)

Times (sec)	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)	Stability Control Yaw Rate (deg/sec)
- 1.4	0.086	-0.045	0.12
- 1.3		-0.031	1.5
- 1.2		-0.647	-1.5
- 1.1		-0.75	-2.5
- 1.0		-0.709	-1.5
- 0.9		-0.763	-1.5
- 0.8		-0.826	-1.5
- 0.7		-0.792	-0.75
- 0.6		-0.895	1.25
- 0.5	0.014	-0.83	1.87
- 0.4	0.03	-0.835	0.75
- 0.3	0.022	-0.607	0.37
- 0.2	0.107	-0.653	0.75
- 0.1	0.05	-0.823	-3.37
0.0	-0.004	-0.811	-3.75

Integrate lat accel - get velocity – integrate again – get displacement

Reaction

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

Polarity on Ford Stability Control

- The steering and yaw rate are opposite the normal right hand rule as documented in SAE 2014-01-0504.
- The paper reported lateral acceleration seemed to be according to right hand rule, based on CAN bus values, but several real world files only make sense if the lateral acceleration is also backwards. In a normal right swerve, all 3 are negative in real world files.

Ford RC7 Data Lims should apply to Ford RC6 and AB10 too

Under Pre-Crash Data -5 to 0 sec

- Steering Wheel Angle if Applicable: positive value indicates left turn, and negative value would indicate right turn.
- Stability Control Lateral Acceleration if Applicable: Lateral Acceleration (Y-direction) is the acceleration along the lateral axis of the vehicle, reported as positive when accelerating to the left.
- Stability Control Longitudinal Acceleration if Applicable: Longitudinal Acceleration (X-direction) is the acceleration along the longitudinal axis of the vehicle, reported as positive when accelerating in a forward direction.
- Stability Control Yaw Rate if Applicable: The Yaw Axis is the vertical axis of the vehicle, generally perpendicular to the plane of the road. A positive Yaw Rate is counter-clockwise when observing the vehicle from above.
- Stability Control Roll Rate if Applicable: The Roll Axis is the longitudinal axis of the vehicle, generally aligned with the primary axis of motion of the vehicle. A positive Roll Rate is counter-clockwise when observing the vehicle from the front.

Ford Occupant Size Classification

Pre-Crash Data -1 sec (First Record)

Ignition cycle, crash	359
Frontal air bag warning lamp, on/off	OFF
Occupant size classification, front passenger (Child size Yes/No [Hex value])	No [$08]
Safety belt status, driver	Driver Buckled

- The Hex Value provides additional information that just child yes/no
- The Decoder is different for Ford's 2 main suppliers, Autoliv and Bosch
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

02011_RCM-AB10P_r001

02007_RCM-RC6_r002

New Second Decoder for Bosch

Autoliv	**Bosch**	
$01	$00	Empty
$02	$01	Child (clearly 55 lbs or less)
$04	$05	Indeterminate (gray zone somewhere between 55 to 110 lbs)
$08	$02	Small Adult (not a child and always registers for 110 lb 5th female adult)
$10	$03	Medium adult (not a 5th female and always registers for what is probably a 50th male adult of 170 lbs)
$20	$04	Large Adult (next bigger threshold than above, 95th male , 223 lbs)

Ford Recent News

- There are timing delays in reporting the precrash data in the early Ford Autoliv RC6 modules (09 F150-Escape-Econoline and many 2010). Use Stability Control System long. accel for timing of accel release and brake on. Delays Fixed in “RC6X”. But we don’t know with certainty when it got fixed on each model. No immediate indication in the CDR file.
- New Autoliv RC7 buffers precrash data every 0.1 sec, “0.0” is less than 0.1 sec old. Better data lims on RC7.

More recent Ford News

- The “Bosch AB10” ACM found in 2012+ Focus and 2015 Escape and other vehicles which have a second event recorded have an issue under review.
- Time between events may say 1.2 seconds, while there is no over lap in the 5 seconds of precrash data, while the overlap in stability control system data is more than 1.2 seconds. Ford fixed new production in Nov. 2015. A disclosure in data limitations is expected sometime.

Available on my website:

Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 9-08-2016, after the original presentation at the Jan 2016 CDR Summit)



New Name: "FCA Group"

FIAT CHRYSLER AUTOMOBILES

Includes Dodge/Jeep/Ram/Lancia

- Chrysler began installing EDR's with good precrash data in some 2005 models.
- Early models recorded only in deployments
- ND recording phased in starting 2009 MY – approaching 563 intent
- Full 563 intent phases in starting 2011



FCA 17.1 CDR Coverage 44%

Last 3 digits of Bosch CDR DTM adpater/cable numbe			2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Chrysler	200	Console, aft of shifter							790/516		387/516		387/785		
Chrysler	300	Center stack		228 [3,5]	228 [5]				387/598						
Chrysler	Aspen	Console, aft of			226 [4]	226									
Chrysler	Cirrus (not sold in US)							387/516 [8]							
Chrysler	Grand Voyager	Center						387/385 [8]							
Chrysler	Pacifica	Center stack			228 [5]										387/785
Chrysler	PT Cruiser	Console, fwd of			228 [5,7]	228 [5]									
Chrysler	Sebring	Console, aft of			226 [4]	226		387/516 [8]							
Chrysler	Town & Country	Center stack				385 [6]		387/385 [8]			387/785				
Dodge	Avenger	Console, aft of				226		387/516 [8]	387/790/516		387/516				
Dodge	Caliber				226 [4]	226		387/516 [8]	387/790/516						
Dodge	Caravan	Center				385 [6]		387/385 [8]			387/785				
Dodge	Challeng	Center stack				228			387/228 [10]	387/228			387/598		
Dodge	Charger	Center stack		228 [3,5]	228 [5]				387/598						
Dodge	Dakota/Raider				228 [5]				228	Mex Only					
Dodge	Dart	Cen									387/785				
Dodge	Durango	Console, aft	226 [2,4]	226 [2,4]	226 [2,4]	226 [2]	226 [2]		387/228 [10]	387/228		387/598			
Dodge	Journey	Console, fwd of shifter					385	387/385 [8]			387/385	387/385 +516PPT			
Dodge	Magnum	Center stack		228 [3,5]	228 [5]										
Dodge	Nitro	Console, aft of			227 [4]	227		387/516 [8]							
Dodge/Ram	Prom City												387/802		
Dodge/Ram	Ram	CF Seat/console		226 [4]				387/453 [8]							
Dodge/Ram	Ram Pickup	Seat/console		226 [4]			226 [9]	387/453 [8]							
Dodge/Ram	Ram Pickup (1500)	nsole		226 [4]		226	453 [8]	387/453 [8]	387/453		387/785				
Dodge/Ram	Ram Van	Cente									387/785				
Dodge/SRT	Viper	Center Stack									387/785				
Jeep	Commander	Console, aft		228 [3,5]	228 [5]										
Jeep	Compass/Patriot	Center stack			226 [4]	226		387/516 [8]	790/516		387/516		387/ *808*		
Jeep	Grand Cherokee	Console, aft of		228 [3,5]	228 [5]				387/228 [1]	387/228+	387/228	387/598			
Jeep	Liberty/Cherokee	Console, aft of				227		387/516 [8]			?	387/785			
Jeep	Renegade													387/819*	
Jeep	Wrangler	Console, aft of			227 [4]	227		387/546 [8]			387/546				

Call FCA

Maybe Call FCA

Frontal Deploy

Frontal/Side Deploy

Non Deploys

563 Non Deploys

Chrysler Data

- Chrysler has been the leader in precrash data with 50 data points at 0.1 sec intervals in most models, but lagged in getting Delta V & some other info
- Key cycle info phased in starting 2010, X and Y Delta V starting 2011, seat belt use and deploy times also phased in, some 2011+ also have odometer readings.
- Some reports of clipping of side accelerometer at 22 or 32g

Chrysler Data every 0.1 seconds!

Pre-Crash Data (Most Recent Event - table 1 of 5)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Vehicle Event Recorder Status	Engine RPM	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, % Full	Accelerator Pedal, % Full	Raw Manifold Pressure (kPa)	Service Brake
-5.0	Complete	4,416	124 [199]	8.7	0.0	15	Off
-4.9	Complete	4,416	124 [199]	8.3	0.0	16	Off
-4.8	Complete	4,416	123 [198]	8.3	0.0	16	Off
-4.7	Complete	4,416	123 [198]	8.3	0.0	16	Off
-4.6	Complete	4,384	122 [197]	8.3	0.0	16	Off
-4.5	Complete	4,384	122 [197]	8.3	0.0	16	Off
-4.4	Complete	4,384	122 [196]	8.3	0.0	16	Off
-4.3	Complete	4,352	122 [196]	8.3	0.0	16	Off
-4.2	Complete	4,352	121 [195]	8.3	0.0	16	Off
-4.1	Complete	4,320	121 [195]	8.3	0.0	16	Off
-4.0	Complete	4,320	121 [194]	8.3	0.0	16	Off
-3.9	Complete	4,320	121 [194]	8.3	0.0	16	Off
-3.8	Complete	4,320	121 [194]	8.3	0.0	16	Off
-3.7	Complete	4,288	120 [193]	8.3	0.0	16	Off

Steering & Yaw Data! (and more)

Pre-Crash Data (Most Recent Event - table 3 of 5)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Steering Input (deg) (if equip.)	Yaw Rate (deg/sec) (if equip.)	Wheel Speed LF (RPM) (if equip.)	Wheel Speed RF (RPM) (if equip.)	Wheel Speed LR (RPM) (if equip.)	Wheel Speed RR (RPM) (if equip.)
-1.5	2	1	1,216	1,214	1,190	1,196
-1.4	-2	1	1,194	1,188	1,170	1,177
-1.3	-2	1	1,158	1,153	1,131	1,141
-1.2	-6	2	1,128	1,095	1,107	1,108
-1.1	-12	2	1,102	1,099	1,060	1,078
-1.0	-20	4	1,033	1,076	1,040	1,057
-0.9	-6	6	982	1,052	[illegible]	1,035
-0.8	32	3	1,017	992	[illegible]	1,014
-0.7	50	-3	989	896	980	986
-0.6	66	-11	984	961	968	953
-0.5	64	-18	968	820	959	902
-0.4	58	-15	950	863	944	921
-0.3	62	-9	922	741	920	890
-0.2	68	-4	840	678	900	873
-0.1	48	-1	847	825	879	855

ABS working

Integrating Yaw Rate

From prior page

- 3 deg/sec * 0.1 sec =	-0.3 degrees
-11 deg/sec * 0.1 sec =	-1.1 degrees
-18 deg/sec * 0.1 sec =	-1.8 degrees
-15 deg/sec * 0.1 sec =	-1.5 degrees
- 9 deg/sec * 0.1 sec =	-0.9 degrees
- 4 deg/sec * 0.1 sec =	-0.4 degrees
- 1 deg/sec * 0.1 sec =	-0.1 degrees
Total Swerve Right	6.1 degrees

Note this is how far nose of car is pointing right – *CG* motion may *lag*

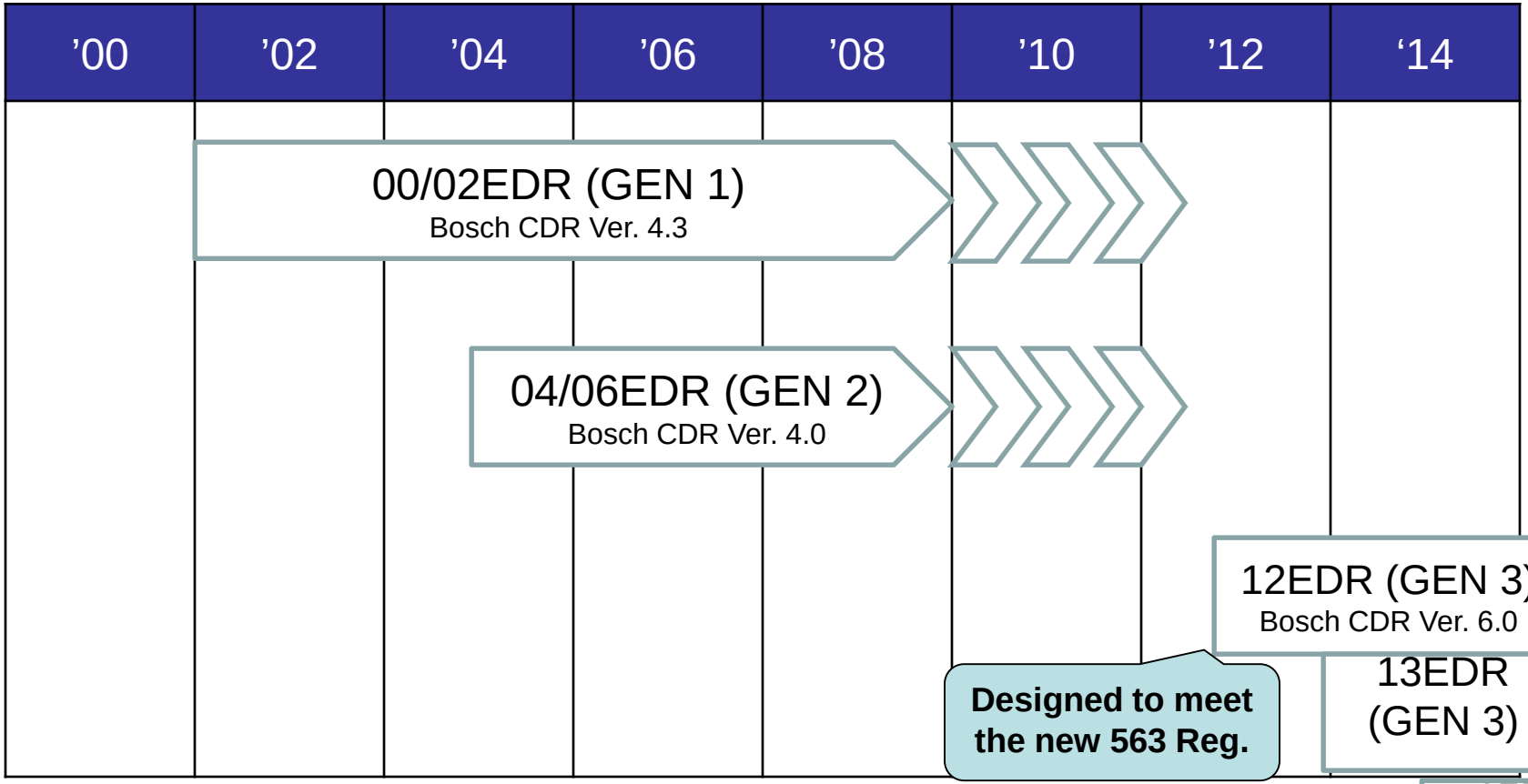
European Pedestrian Protection

- Starting 2014, In Europe only at this time, Chrysler has added a pedestrian protection feature to the Dodge Journey, Fiat Freemont, and Lancia Grand Voyager to deploy a feature that pops the hood up slightly to make the angle better and less injurious to pedestrians.
- It is a separate module with separate data and must be imaged separately thru the DLC or via cable 516.
- It contains useful to ped crash investigations

Other FCA News

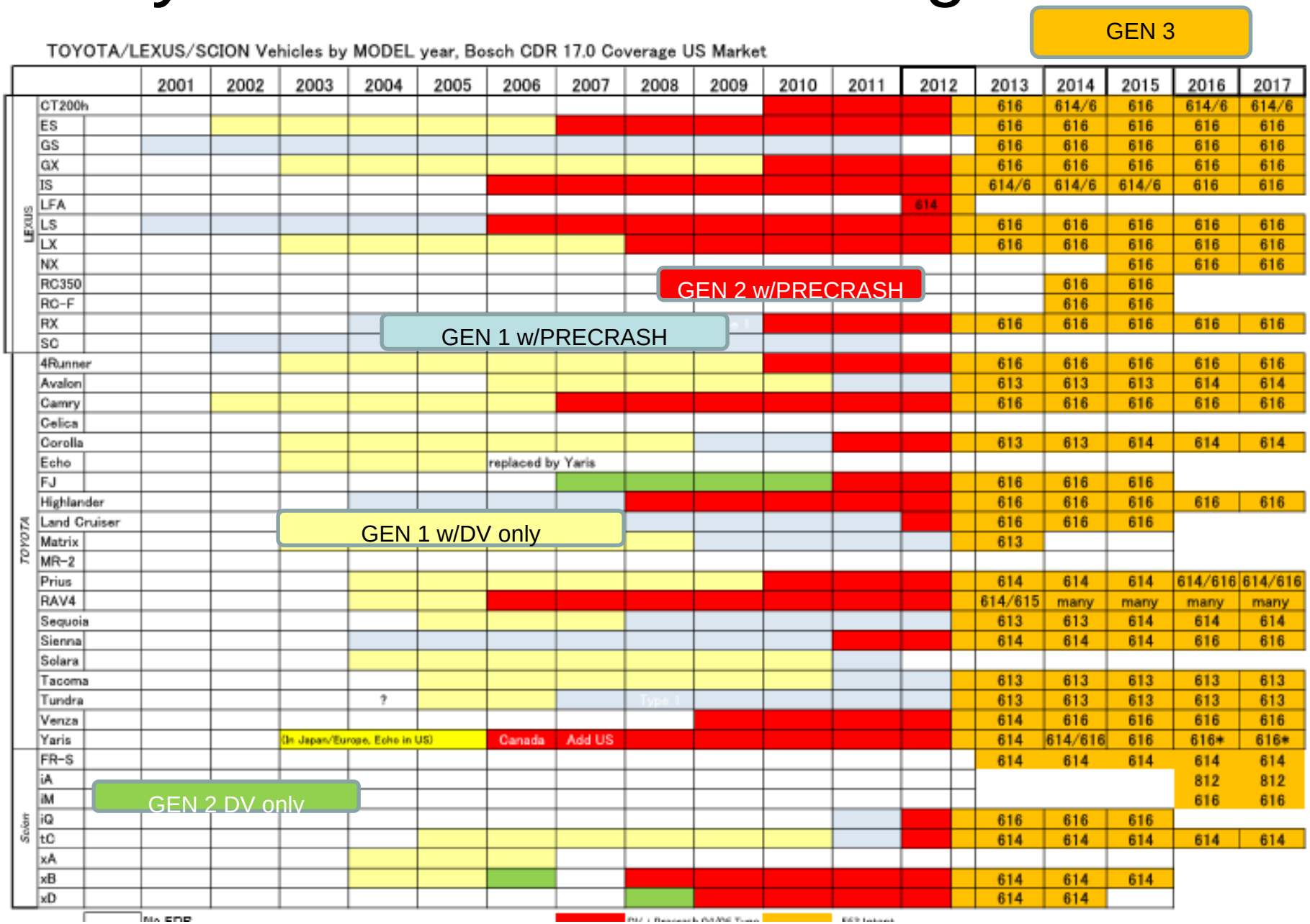
- FCA reviewed may past entries, have fixed ***many*** small things, fixed polarities, fixed incorrect module locations, revised reporting of deploy times (was stage 1 to stage 2 vs AE to stage 2).
- 10-12 Journey + Minivan – all ND's display as "interrupted" recording
- 2013 TRW – key cycle counters may not match perfectly (different sources)
- Most "features" are noted in data limitations. i.e recordings under 5mph DV

History of Toyota EDR



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Toyota CDR 17.1 Coverage 74%



		2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
LEXUS	CT200h													616	614/6	616	614/6	614/6
	ES													616	616	616	616	616
	GS													616	616	616	616	616
	GX													616	616	616	616	616
	IS													614/6	614/6	614/6	616	616
	LFA												614					
	LS													616	616	616	616	616
	LX													616	616	616	616	616
	NX															616	616	616
	RC350														616	616		
	RC-F														616	616		
	RX													616	616	616	616	616
	SC																	
TOYOTA	4Runner													616	616	616	616	616
	Avalon													613	613	613	614	614
	Camry													616	616	616	616	616
	Celica																	
	Corolla													613	613	614	614	614
	Echo																	
	FJ													616	616	616		
	Highlander													616	616	616	616	616
	Land Cruiser													616	616	616		
	Matrix													613				
	MR-2																	
	Prius													614	614	614	614/616	614/616
	RAV4													614/615	many	many	many	many
	Sequoia													613	613	614	614	614
	Sienna													614	614	614	616	616
	Solara																	
	Tacoma													613	613	613	613	613
	Tundra													613	613	613	613	613
	Venza													614	616	616	616	616
	Yaris													614	614/616	616	616*	616*
Scion	FR-S													614	614	614	614	614
	iA																812	812
	iM																616	616
	iQ													616	616	616		
	tC													614	614	614	614	614
	xA																	
	xB													614	614	614		
	xD													614	614			

Toyota Gen 1 & 2 Precrash

Extra data point!

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 2)

Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	57.2 [92]	55.9 [90]	52.2 [84]	52.2 [84]	38.5 [62]	0 [0]
Brake Switch	OFF	ON	OFF	ON	ON	ON
Accelerator Rate (V)	0.94	0.78	0.78	0.78	0.86	0.78
Engine RPM (RPM)	2,400	2,000	1,600	1,600	800	400

Volts

But 0.5 sec old!

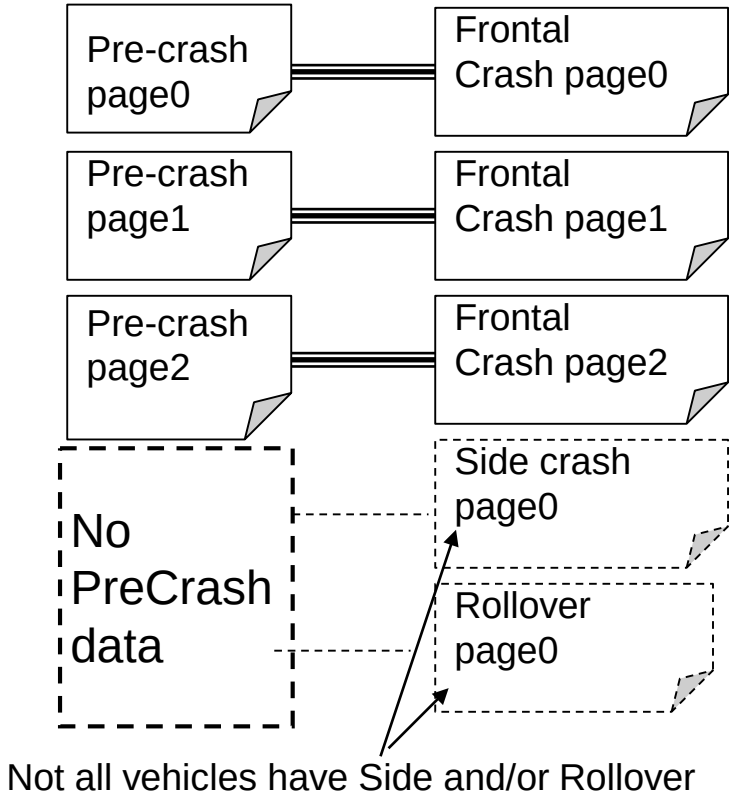
75 mph max speed

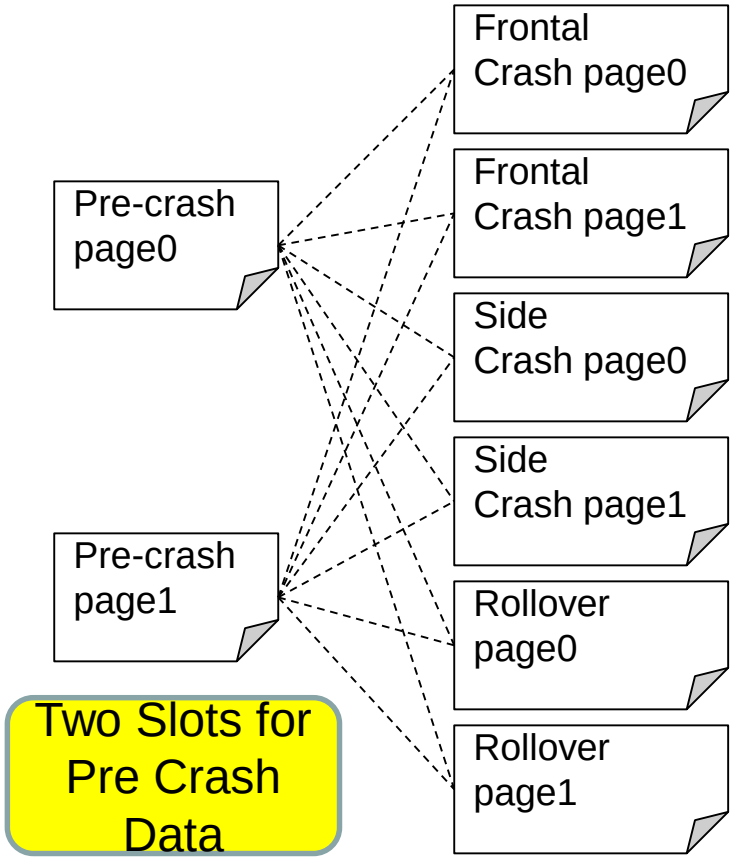
Toyota Gen 3 –”12 EDR”

- Adds key cycles, deploy times, longer ΔV (250ms), more pre-crash 5 sec @ 0.5 sec intervals, top speed to 122 mph (200 kph) vs 75 or 78 mph in earlier models
- Accel pedal in % vs voltage.
- Frontal Recording threshold 5mph ΔV min.
- Lat ΔV only recorded in side wakeups and for only 68-72ms after wakeup on satellites
- Throttle %, Steering, Yaw Rate, Brake Pressure, VSC Long Accel, Cruise, Gear, RPM @ 100 (vs 400’s).

Toyota EDR "Pages" in memory

00/02EDR (GEN 1)	04/06EDR (GEN 2) and **12EDR (GEN 3)**
▪Frontal Crash: **3** pages ▪Side and Rollover: 1page (no Pre-Crash)	▪Pre-Crash : **2** pages ▪Frontal Side Rollover :2 pages each





Six Slots for Crash Data

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Unique to Toyota

- Frontal data if front algorithm wakes up
- Side data if side algorithm wakes up
- Side Delta V data for about 70ms may not be enough to capture crash pulse

Toyota 12 EDR Makes Big Improvements

Parameter	04/06 EDR		12 EDR	
Vehicle Speed (Part 563)	2 km/h /LSB	0~122 [km/h]	1 km/h /LSB	0 ~ 200 [km/h]
Accelerator Pedal (Part 563)	.039 V/LSB	0~5 [V]	0.5 %/LSB	0 ~ 100 [%]
Engine RPM (Part 563)	400 RPM/LSB	0~5200 [RPM]	100 RPM/LSB	0 ~ 12800 [RPM]
Service Brake (Part 563)	ON/OFF		ON/OFF	
Shift Position	R / N/ P / D		R / N / P / D / 5 / 4/ 3 / 2/ LO/ B/ Seq	
Percentage of Engine Throttle			0.5%/LSB	0 ~ 100 [%]
Motor RPM (Part 563)			100 RPM/LSB	-7500 ~ 17500 [RPM]
Brake Oil Pressure			0.048 MPa	0 ~ 12.144 [MPa]
Sequential Shift			1 / 2 / 3 /4 / 5 / 6 / 7 / 8	
Steering Input (Part 563)			3 deg/LSB	-375 ~ 375 [deg]
VSC Yaw Rate			0.488 deg/s/LSB	-61 ~ 61 [deg/s]
VSC Longitudinal Acceleration			0.07178 m/s^2/LSB	-8.9725~8.9725 [m/s^2]
Cruise Control Status			ON/OFF	
Drive Mode - Power Drive Mode – Eco Drive Mode - Sport Drive Mode - Snow Drive Mode - EV			ON/OFF	

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12 EDR Precrash Data Table

Time (sec)	-4.8	-4.3	-3.8	-3.3	-2.8
Vehicle Speed (MPH [km/h])	0 [0]	0.6 [1]	1.9 [3]	5 [8]	8.1 [13]
Accelerator Pedal, % Full (%)	8.0	7.5	30.5	79.0	100.0
Percentage of Engine Throttle (%)	0.5	0.5	3.0	17.5	26.0
Engine RPM (RPM)	700	700	700	1,700	1,600
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF
Brake Oil Pressure (Mpa)	DIVIDE BY 12, get %		0.00	0.00	0.00
Longitudinal Acceleration - VSC	DIVIDE BY 9.8, get G's		-1.148	-3.087	-2.943
Yaw Rate (deg/sec)	0.00	2.44	6.34	14.15	20.01
Steering Input (degrees)	-246	-255	-318	-273	-246
Shift Position	R	R	R	R	R

Toyota 12 EDR Drawback

- Lateral acceleration is only recorded in a side algorithm wakeup.
- When it is recorded, it is only for 68ms
- It is only recorded for the side satellites, not for the central control module
- If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location

New “13 EDR” & “15 EDR”

- 13 EDR Adds Lateral Accel data from ACM in frontal events for 200ms (Fixes biggest drawback to 12 EDR)

Longitudinal/Lateral Crash Pulse (Most Recent Event

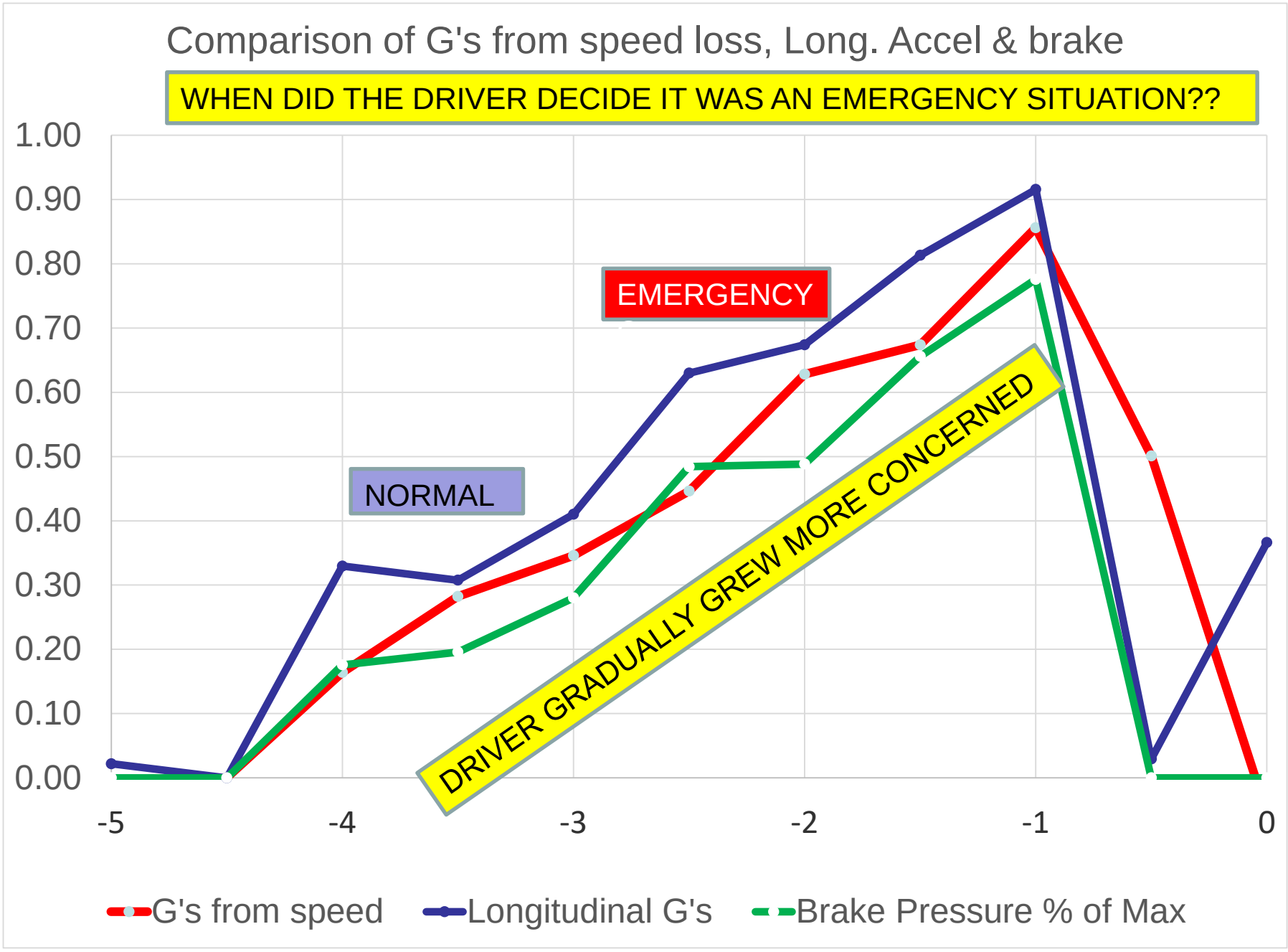
Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	F
0	0.0 [0.0]	0.0	
10	0.3 [0.6]	-1.9	
20	0.5 [0.8]	0.0	

- Calculate Lat Delta V for each 10ms interval
- DV_{10ms} = Accel/9.8*(32.2/1.466)*0.01s
- Add up all the intervals to get the Delta V

3 ways to get braking rate

- Take consecutive speed data points, subtract them to get speed change vs time, divide by accel due to gravity to get how many G's. For -1.5 sec:

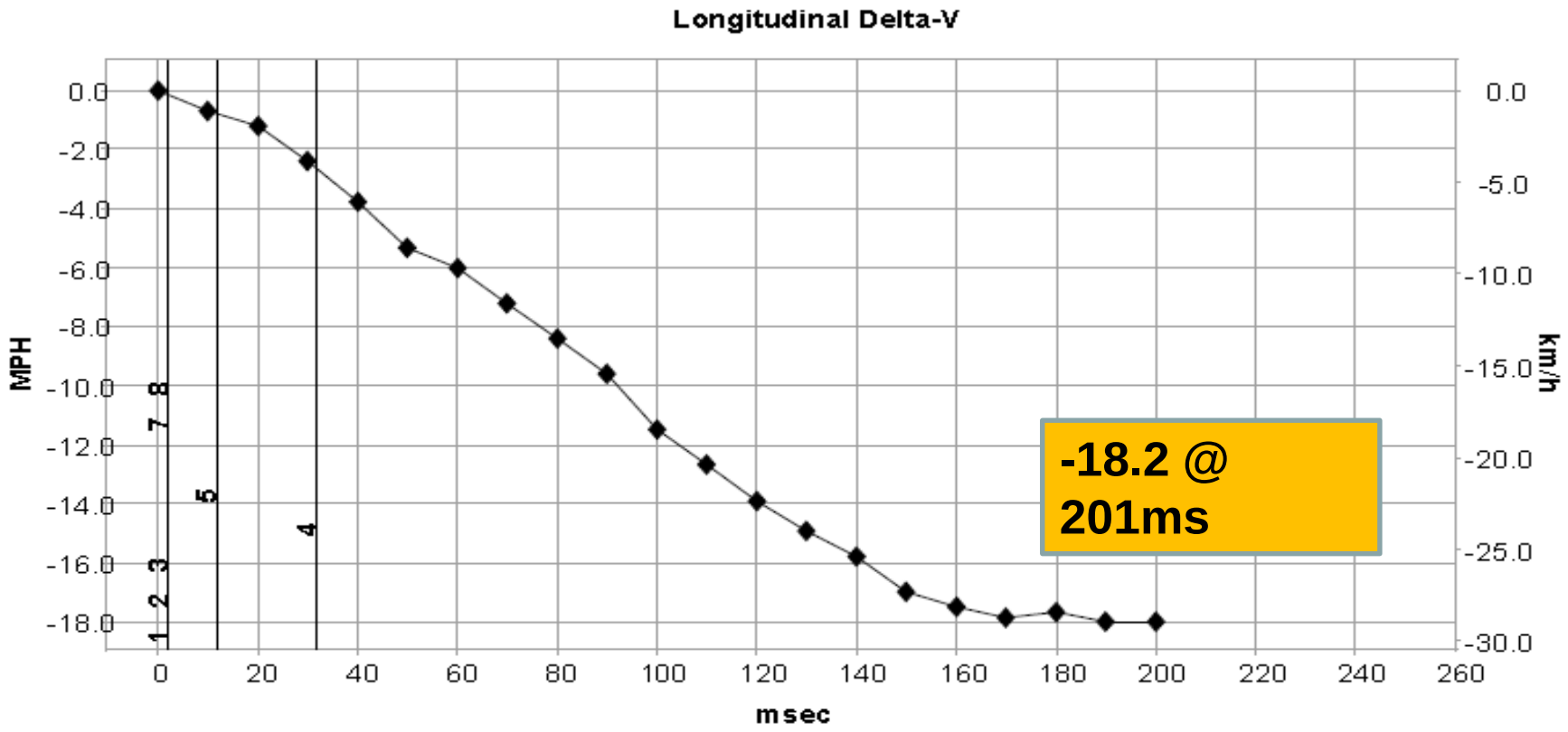
  (29.8-22.4) = 7.4 mph*1.466/32.2/0.5 sec = 0.67G
- Compare this to longitudinal acceleration in m/s, divide by accel due to gravity of 9.8 m/sec^2 to get G's -7.968m/sec^2 / 9.8 = 0.82G
- Compare this to % max brake pressure in Mpa 7.87/12 = 66%
- See next slide for graph comparing 3 methods


Comparison of G's from speed loss, Long. Accel & brake
WHEN DID THE DRIVER DECIDE IT WAS AN EMERGENCY SITUATION??
1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
-5
-4
-3
-2
-1
0
EMERGENCY
NORMAL
DRIVER GRADUALLY GREW MORE CONCERNED
G's from speed
Longitudinal G's
Brake Pressure % of Max

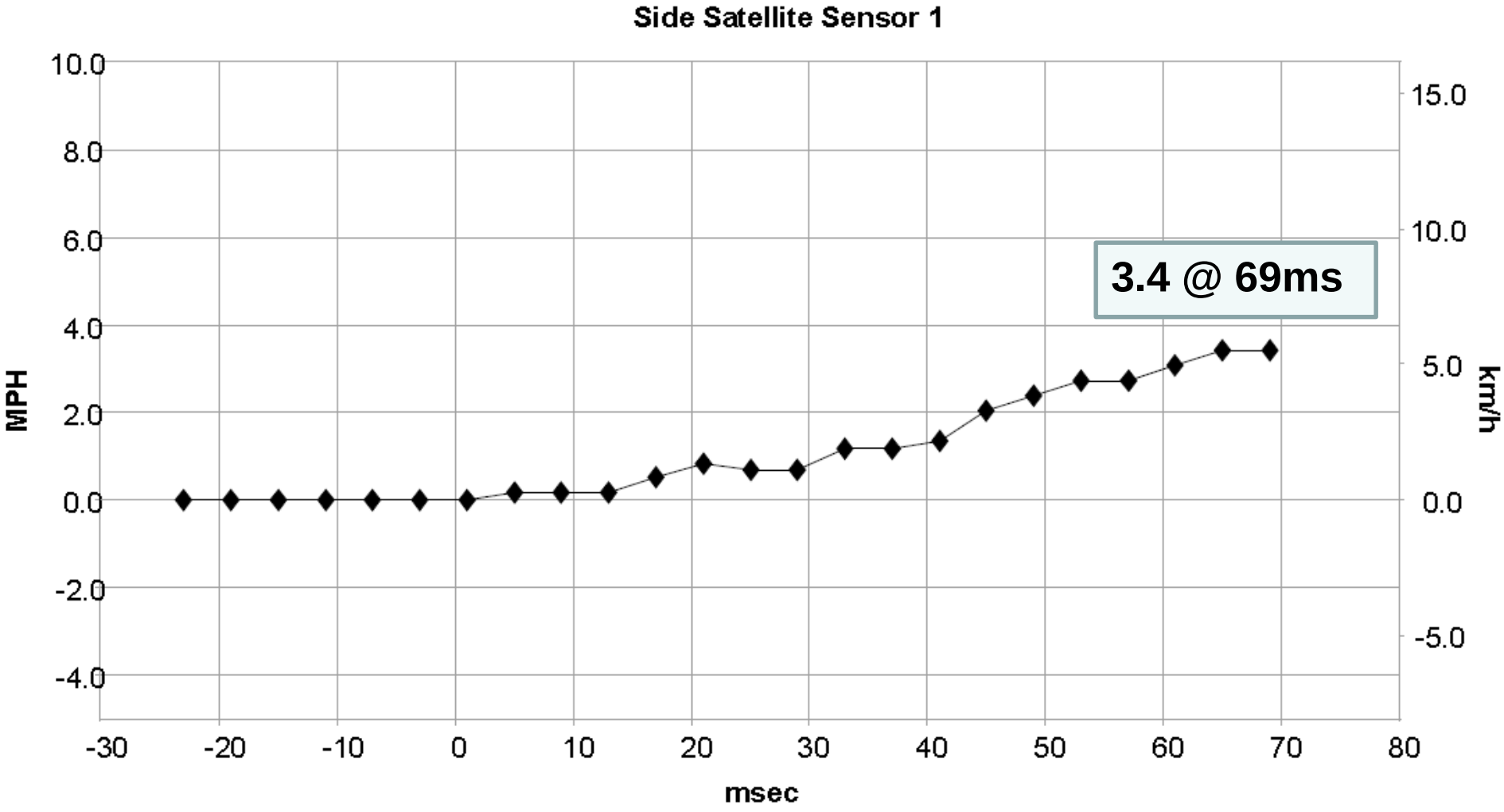
Delta V analysis – What's the PDOF? The Total??

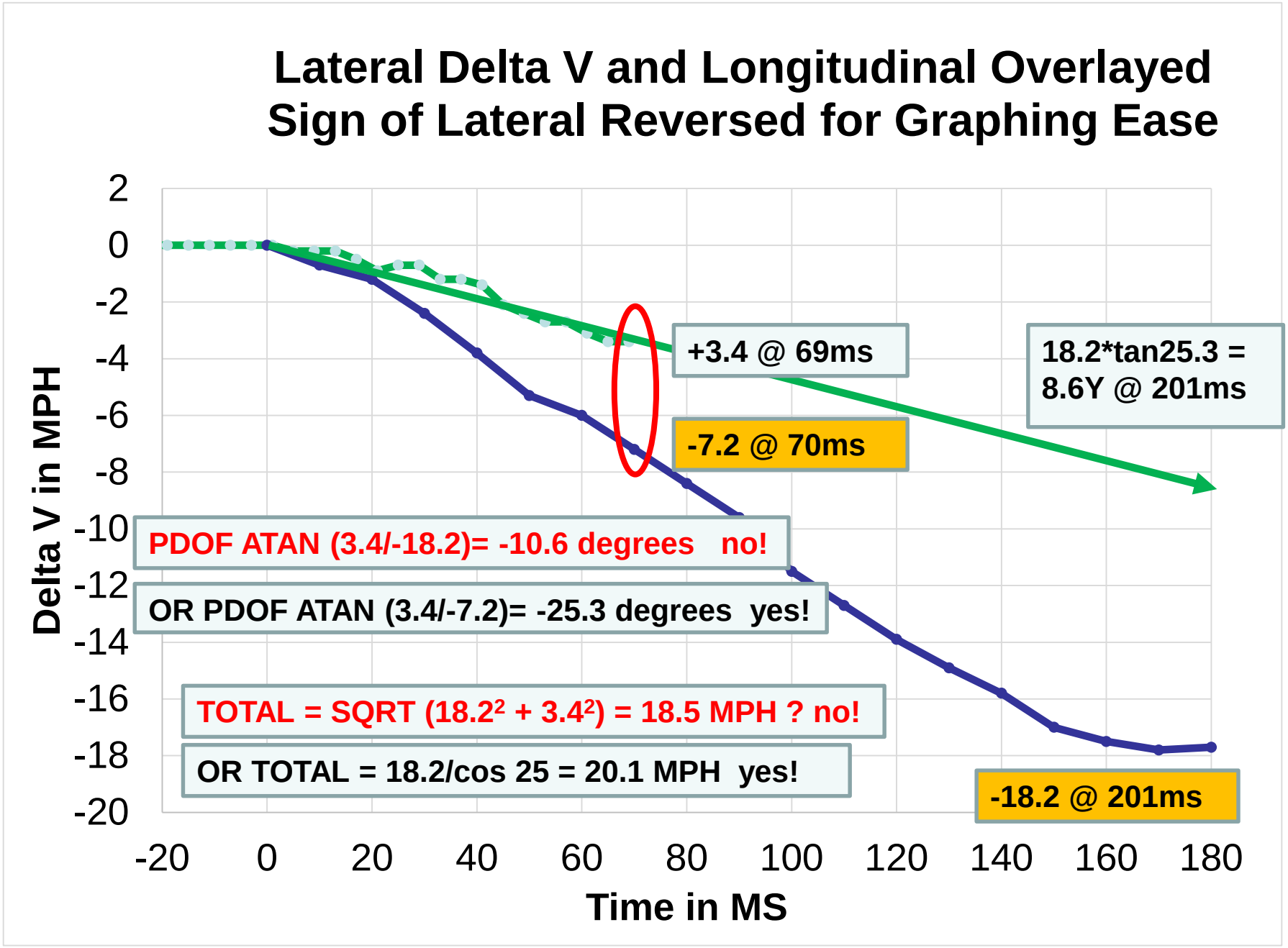
Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	2.0
Length of Delta-V (msec)	200
Max. Longitudinal Delta-V (MPH [km/h])	-18.2 [-29.2]
Time, Maximum Delta-V, Longitudinal (msec)	201.0
Power Supply Status at Max. Delta-V	ON



Lateral Delta V -




Lateral Delta V and Longitudinal Overlayed
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees yes!
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

New “Vehicle History Utility”

- New feature in 2013/4 Rav 4 and 2014 Highlander, IS250/250, 2015 Prius, NX200t, NX300h, RC350, RCF
- Triggers EDR-style recording (5-8 secs pre and post) if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion (26 triggers)
- Accessible only with Toyota Techstream service tool – **not** part of Bosch CDR system
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a pattern of aggressive driver behavior. Odometer included. Great for Road Rage cases.
- Access this whenever EDR data is either ***not available*** or ***insufficient*** for your investigation

NISSAN EDR's (CDR 9/1/12+, 24%)

Model	Location	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR cable number													
INFINITI													
G CONVERTIBLE	TUNNEL BETWEEN SEATS				*		Jul-12		780				
G COUPE	TUNNEL BETWEEN SEATS		*		Yellow = Proprietary Consult Tool Only				780				
G SEDAN	TUNNEL BETWEEN SEATS		*	No direct to module cables			Jul-12		780				
M	TUNNEL BETWEEN SEATS								780				
Q40	TUNNEL BETWEEN SEATS										780		
Q50	TUNNEL BETWEEN SEATS								780	780	780	780	780
Q60	TUNNEL BETWEEN SEATS								780	780	780		780
Q70	TUNNEL BETWEEN SEATS								780	780	780	780	780
EX	TUNNEL BETWEEN SEATS			Dealers do NOT have special software					780				
FX	TUNNEL BETWEEN SEATS			required to read EDR					780				
JX35	TUNNEL BETWEEN SEATS								780				
QX50-80	TUNNEL BETWEEN SEATS								780	780	780	780	780
Q30, QX30													780
NISSAN		2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
370Z	TUNNEL BETWEEN SEATS		*				Aug-12		780	780	780	780	780
ALTIMA COUPE	TUNNEL BETWEEN SEATS			*					780				
ALTIMA SEDAN	TUNNEL BETWEEN SEATS		*	Yellow = Proprietary Consult Tool Only					780	780	780	780	780
ARMADA	TUNNEL BETWEEN SEATS	*		No direct to module cables					780	780	780		780
CUBE	TUNNEL BETWEEN SEATS				*				780	780			
FRONTIER	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780	780	780
GT-R	TUNNEL BETWEEN SEATS				*					780	780	780	780
JUKE	TUNNEL BETWEEN SEATS								780	780	780	780	780
LEAF	TUNNEL BETWEEN SEATS								780	780	780	780	780
MAXIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780
MICRA	Check Manual										780	780	780
MURANO	TUNNEL BETWEEN SEATS		*		*				780	780	780	780	780
MURANO CROSS C	TUNNEL BETWEEN SEATS								Included in above				
NV200	TUNNEL BETWEEN SEATS								780	780	780	780	780
PATHFINDER	TUNNEL BETWEEN SEATS	*							780	780	780	780	780
QUEST	CTR CONSOLE NEAR FRONT		*						780	780	780	780	780
ROGUE	Check Manual			*					780	780	780	780	780
SENTRA	TUNNEL BETWEEN SEATS		*						780	780	780	780	780
TITAN	TUNNEL BETWEEN SEATS	*							780	780	780	780	780
VERSA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	
VERSA NOTE HB	TUNNEL BETWEEN SEATS						Sep-12		780	780	780	780	780
XTERRA	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780		

Yellow area not covered by CDR

563

Nissan Pre-Crash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % full	Engine RPM	Motor RPM	Service Brake (On, Off)	Steering Input (deg)
-5.0	24 [39]	16	2650	0	Off (Brake Not Activated)	-32
-4.5	23 [37]	11.5	1750	0	Off (Brake Not Activated)	-2
-4.0	21 [34]	9	1200	0	On (Brake Activated)	28
-3.5	15 [24]	9	1150	0	On (Brake Activated)	14
-3.0	13 [21]	4	950	0	On (Brake Activated)	4
-2.5	11 [17]	4	950	0	On (Brake Activated)	4
-2.0	5 [8]	4	900	0	On (Brake Activated)	-48
-1.5	3 [5]	4	850	0	On (Brake Activated)	-98
-1.0	1 [2]	4	850	0	On (Brake Activated)	-160
-0.5	3 [5]	10.5	1100	0	Off (Brake Not Activated)	-194
0.0	6 [9]	16	1150	0	Off (Brake Not Activated)	-218

Hybrid/Electric

Table 2 – not rq'd

At AE

Nissan Polarity Issues Fixed

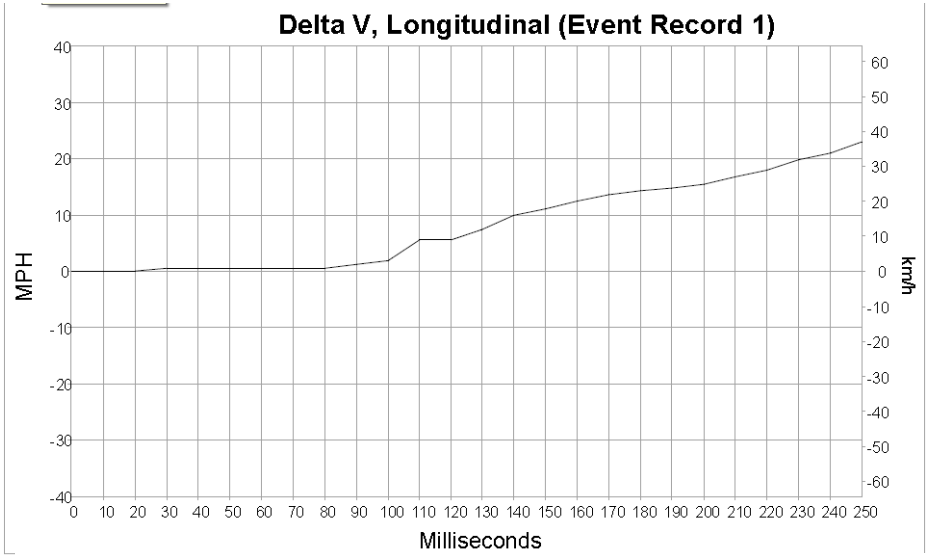
Some exceptions still reported

2013 Altima in CDR 12.2

System Status at Event (Event Record 1)

Frontal Air Bag Suppression Switch Status	S airbag Inhibit)
Maximum Delta-V, Longitudinal (MPH [km/h])	29 [47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

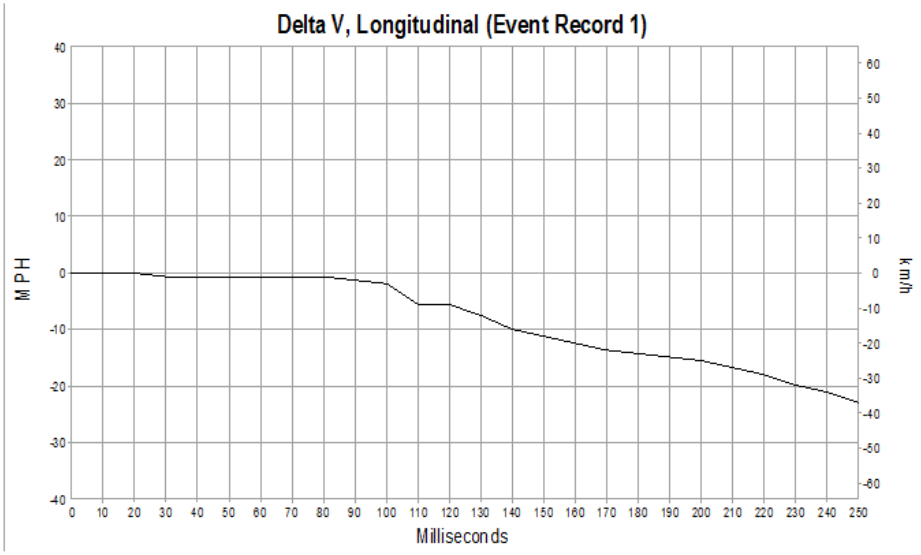
+29 at 300ms



2013 Altima in CDR 16.5

S airbag Inhibit)	
Maximum Delta-V, Longitudinal (MPH [km/h])	-29 [-47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	-32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

-29 at 300ms



Mazda CDR 17.1 Coverage **25%**

Model	Location									
Last 3 digits of Bosch CDR cable number shown		2009	2010	2011	2012	2013	2014	2015	2016	2017
MAZDA				All Mazda use 387 adapter for direct to module						
CX-3	TUNNEL BELOW IP								812	
CX-5	TUNNEL AFT SHIFTER					779				
CX-9	TUNNEL BELOW IP			778					811	
Mazda 2	TUNNEL AFT SHIFTER			792				not liste	812	
Mazda 3	TUNNEL AFT SHIFTER				779				811	
Mazda 5	TUNNEL BELOW IP				778					
Mazda 6	TUNNEL BELOW IP			792			779		811	
MX-5 Miatta	TUNNEL BELOW IP					no coverage			812	

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers. Mostly Delta V.

Typical Mazda Precrash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, % full	Service Brake (On, Off)
-5.0	19 [31]	0	On
-4.5	17 [27]	0	On
-4.0	14 [23]	0	On
-3.5	12 [19]	0	On
-3.0	11 [17]	25	Off
-2.5	10 [16]	61	Off
-2.0	11 [17]	63	Off
-1.5	13 [21]	64	Off
-1.0	16 [26]	64	Off
-0.5	17 [27]	0	Off
0.0	4 [6]	0	Off

Minimum Regulatory Content

This is a second event

Suzuki CDR 17.1 Coverage 13%

Model	Location							
Last 3 digits of Bosch CDR cable number shown		Earlier Models	2011	2012	2013	2014	2015	2016
SUZUKI								
Equator	TUNNEL BETWEEN SEATS	Nissan	Frontier	Sister	387/780			
Grand Vitara	UNDER CENTER CONSOLE				387/793	387/793	Suzuki left the	
Kizashi	UNDER CENTER CONSOLE				387/795	387/795	US Market,	
SX4	UNDER CENTER CONSOLE				387/794	387/794	not released	
XL7 (GM clone)	CENTER TUNNEL	2007-2010						
Swift		1995-2001						
Sidekick		1996-1998						
Vitara		1999-2004						

Suzuki Left the US Market in 2013 – few if any cars here

Volvo CDR 17.1 Coverage 8%

Model	Location	2010	2011	2012	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR cable number sho									
VOLVO		All Volvo use 387 Adapter for DTM images							
S60	CTR TUNNEL UNDER CONSOLE		799						
S80, S80L	CTR TUNNEL UNDER CONSOLE				799				
S90	CTR TUNNEL UNDER CONSOLE								816
V40	CTR TUNNEL UNDER CONSOLE								Europe Onl
V60	CTR TUNNEL UNDER CONSOLE		799						
V70	CTR TUNNEL UNDER CONSOLE				799				
V90	CTR TUNNEL UNDER CONSOLE								816
XC60	CTR TUNNEL UNDER CONSOLE				799				
XC70	CTR TUNNEL UNDER CONSOLE				799				
XC90	CTR TUNNEL UNDER CONSOLE							816	

Old Volvo DV only Law Enf only call Volvo

Volvo Sample Pre-Crash Data

Pre-Crash -5 to 0 sec (Event Record 1)

Mostly minimum to meet 563

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, Percent Full (%)	Service Brake (On, Off)
-5.0	3.1 [5.0]	0.0	Off
-4.5	3.1 [5.0]	0.0	Off
-4.0	3.1 [5.0]	0.0	Off
-3.5	5.0 [8.0]	1.0	On
-3.0	8.1 [13.0]	3.0	Off
-2.5	11.2 [18.0]	5.0	Off
-2.0	14.3 [23.0]	7.0	Off
-1.5	17.4 [28.0]	9.0	On
-1.0	20.5 [33.0]	11.0	On
-0.5	23.6 [38.0]	13.0	On
0.0	26.7 [43.0]	15.0	Off

No RPM

No Steering

Some reports of last data point taken after AE received

Volvo Experience

- 2011 Volvo S60 staged collision at PSP
- Baseline showed 2 ND events labeled as D's – but nothing deployed and they were not locked
- New D event overwrote the earlier ND (that was labeled as a deployment)
- Delta V polarity may be opposite of SAE

BMW CDR 17.1 Coverage 19%

See help file for Rolls Royce

Model	Location	2012	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR adapter/cable number shown							
BMW	**All BMW require 387 Adapter for DTM**						
1 Series					not covered		
2 Series	Driver side, behind IP, rt of strg			796			
3 Series	Driver side, behind IP, rt of strg		796				
Active Hybrid 3	Passenger side, behind the glove box		796				
4 Series	Driver side, behind IP, rt of strg			796			
5 Series	Passenger side, behind the glove box		796				
6 Series	Passenger side, behind the glove box		796				
7 Series	Passenger side, behind the glove box		796			500/796	500/822
X3 xDrive	Passenger side, behind the glove box		796				
X5	Center Tunnel, Between Seats			796			
X1	Center Tunnel, Between Seats				798	500/798	500/807
X3, X4, X6	See Help File - Various				796		
Z4	Center Tunnel, Under Dash				798		
i3, i8	See Service Manual			500/807			

The source data for this chart is the Bosch CDR help file for Version 17.0

Not all models covered

No “back model” coverage

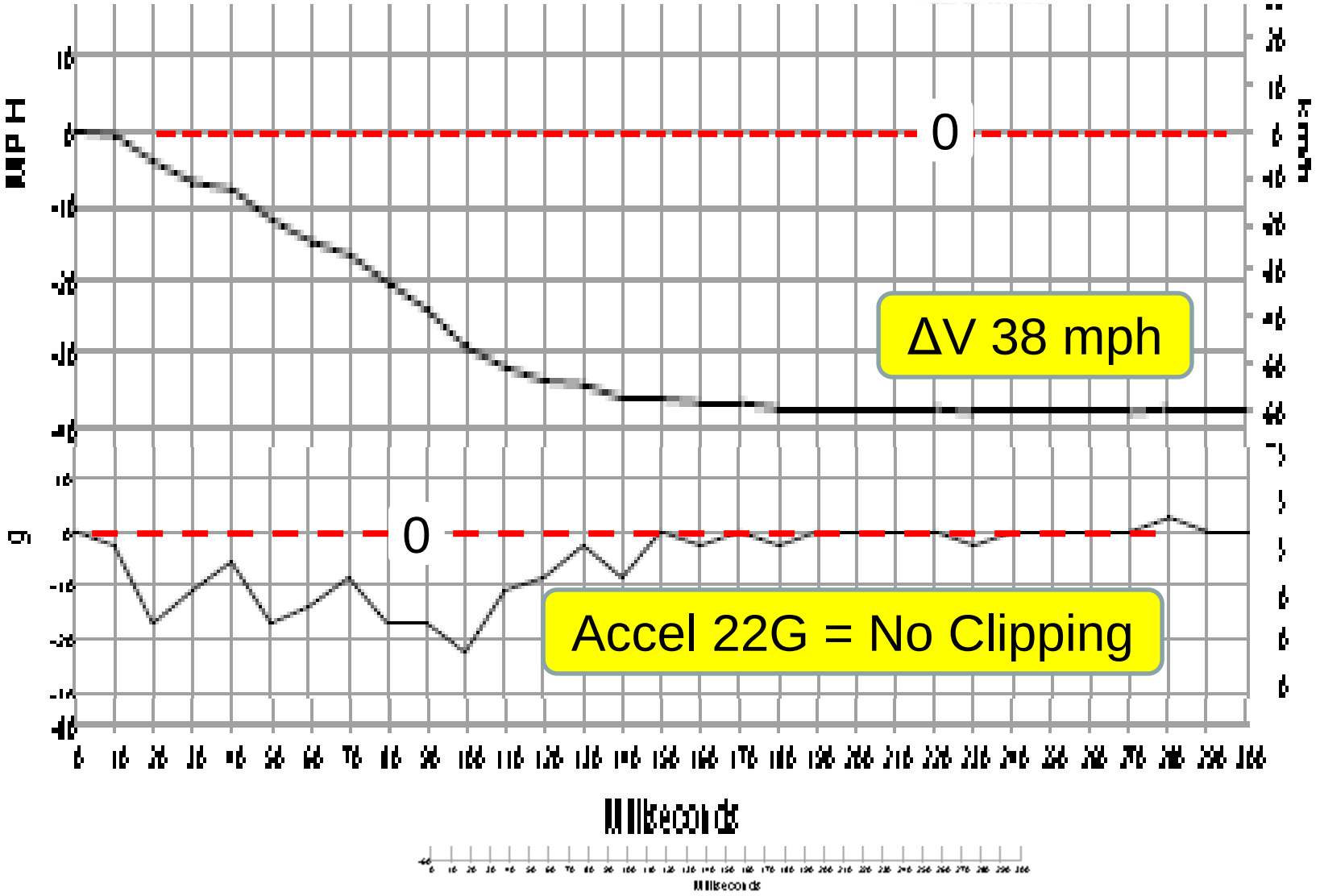
flex ray

BMW Sample Pre-Crash

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Engine RPM	Steering Input (deg)	Service Brake, On/Off	ABS Activity (Engaged, Non-engaged)	Stability Control (On Engaged, Non-engaged)
-5.0	55 [89]	16	1500	0	Off	No ABS Activity	Non-engaged
-4.5	55 [89]	17	1500	0	Off	No ABS Activity	Non-engaged
-4.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.5	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-2.5	55 [89]	8	1500	0	Off	No ABS Activity	Non-engaged
-2.0	55 [89]	7	1500	0	Off	No ABS Activity	Non-engaged
-1.5	55 [89]	14	1500	0	Off	No ABS Activity	Non-engaged
-1.0	55 [88]	19	1500	0	Off	No ABS Activity	Non-engaged
-0.5	55 [89]	0	1500	0	Off	No ABS Activity	Non-engaged
0.0	45 [73]	0	1300	10	On	ABS Activity	Non-engaged

BMW – Both ΔV & Accel



Recent BMW news

- Many updates to software – be SURE to use latest software to read and interpret – DLC was not working on some models due to gateway module issue
- Lateral DV polarity MAY be inverted on 2014 Mini, BMW i3/i3rex, i8 – if car was serviced after 3/2015 it may be fixed.
- OCS “features” under investigation
- Steering polarity recently reversed on some models

Mercedes 17.1 Nov 4%

Model	Start of Sales	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR DTM cable number shown						
Mercedes	EDR Start	All Mercedes use 387 adapter for direct to module downloads				
AMG-GT					800	
B-Class EV	after 4/1/2014			598		
B-Class	after 6/2/2014			598		
C Class Sedan	after 9/1/2014			801		
C Class Coupe	start of 2015 M.Y.			800	801	
CLA Class	after 6/2/2014			598		
CLS-Class				800		
E Class (includes coupe)	after 4/1/2013		800			
G Class	after 11/1/2014			387/598		
GL Class	after 7/11/2014			598	?	?
GLA Class	after 6/2/2014			598		
GLC Class					801	
GLE Class					598	
GLK	after 12/2013		800		?	?
GLS Class					598	
Maybach S600					801	
ML-Class	after 7/11/2014			598	?	?
S-class Sedan	after 9/2013		801			
S-Class Coupe	after 9/1/2014			801		
SL	after 12/2013		800			
SLK	after12/2013		800			
Vito					598	
Smart 2015 & 2016 with "E"	after 9/1/2014			809		
Smart 2016 with VIN 4th "F" &2017					820	820

Mercedes Details

- Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
- Precrash speed, brake, throttle & longitudinal ΔV, 563 table 1 items only.
- Can hold 5 events, D's lock, ND's FIFO
- Recording threshold is 5 mph over 150ms
- Speed is avg. of 4 wheel speed sensors

Mercedes Pre Crash Data

Minimum required by law

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal (%)	Service Brake Activation
-5.0	48 [77]	0	On
-4.5	47 [76]	16	Off
-4.0	48 [77]	15	Off
-3.5	48 [77]	20	Off
-3.0	48 [77]	20	Off
-2.5	48 [78]	21	Off
-2.0	49 [79]	0	On
-1.5	46 [74]	0	On
-1.0	39 [63]	0	On
-0.5	31 [50]	0	On
0.0	25 [40]	0	On

Any EDR data in older Mercedes? NO

- Some diagnostic codes may include “freeze frame” data, but airbag codes don’t record speed. Some codes do.
- By the time those other codes get set after a crash, the speed is no longer the precrash speed.
- Mercedes will assist law enforcement. Call 1-88-4Mercedes.

Sample Mercedes File

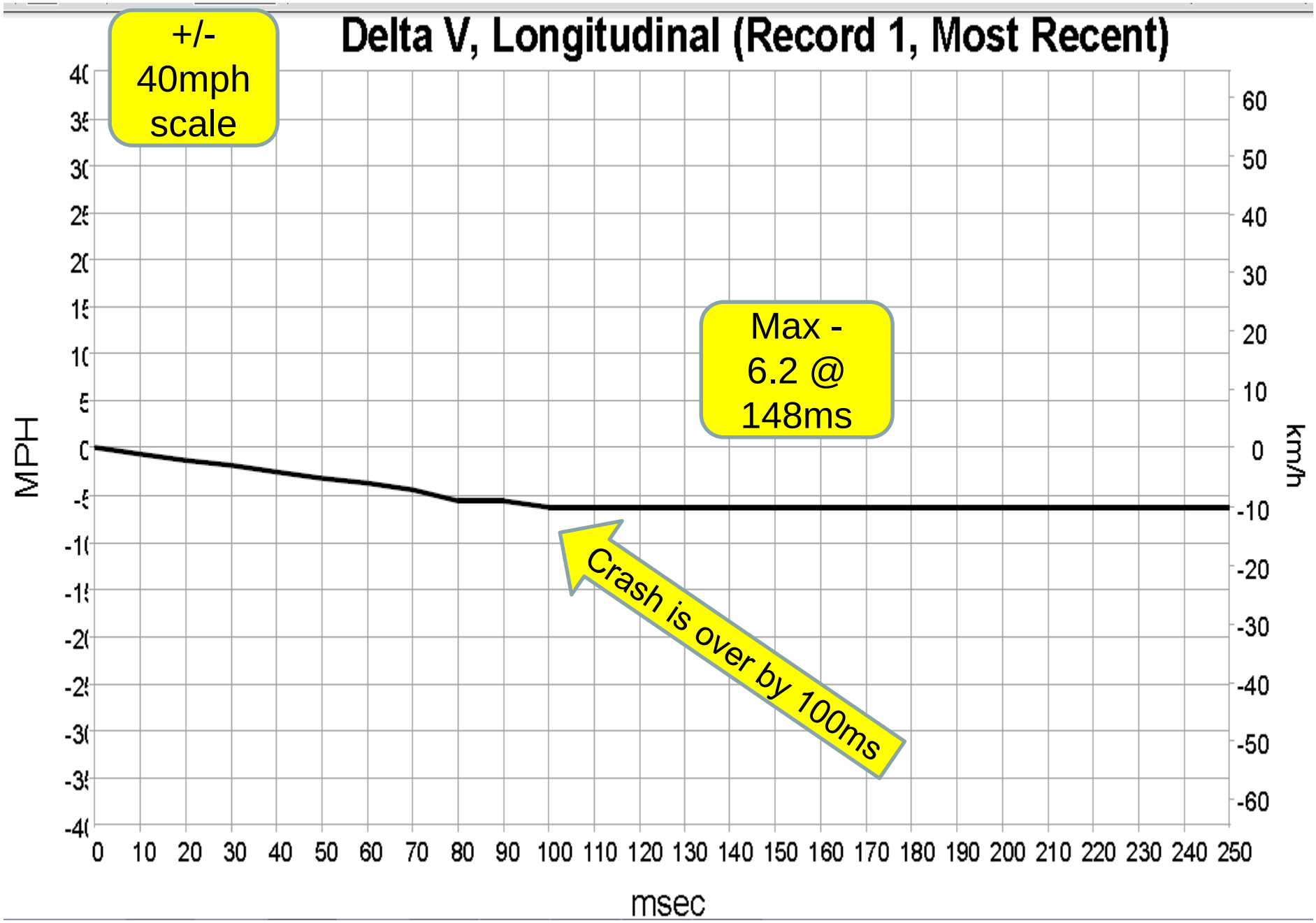
System Status at Event (Record 1, Most Recent)

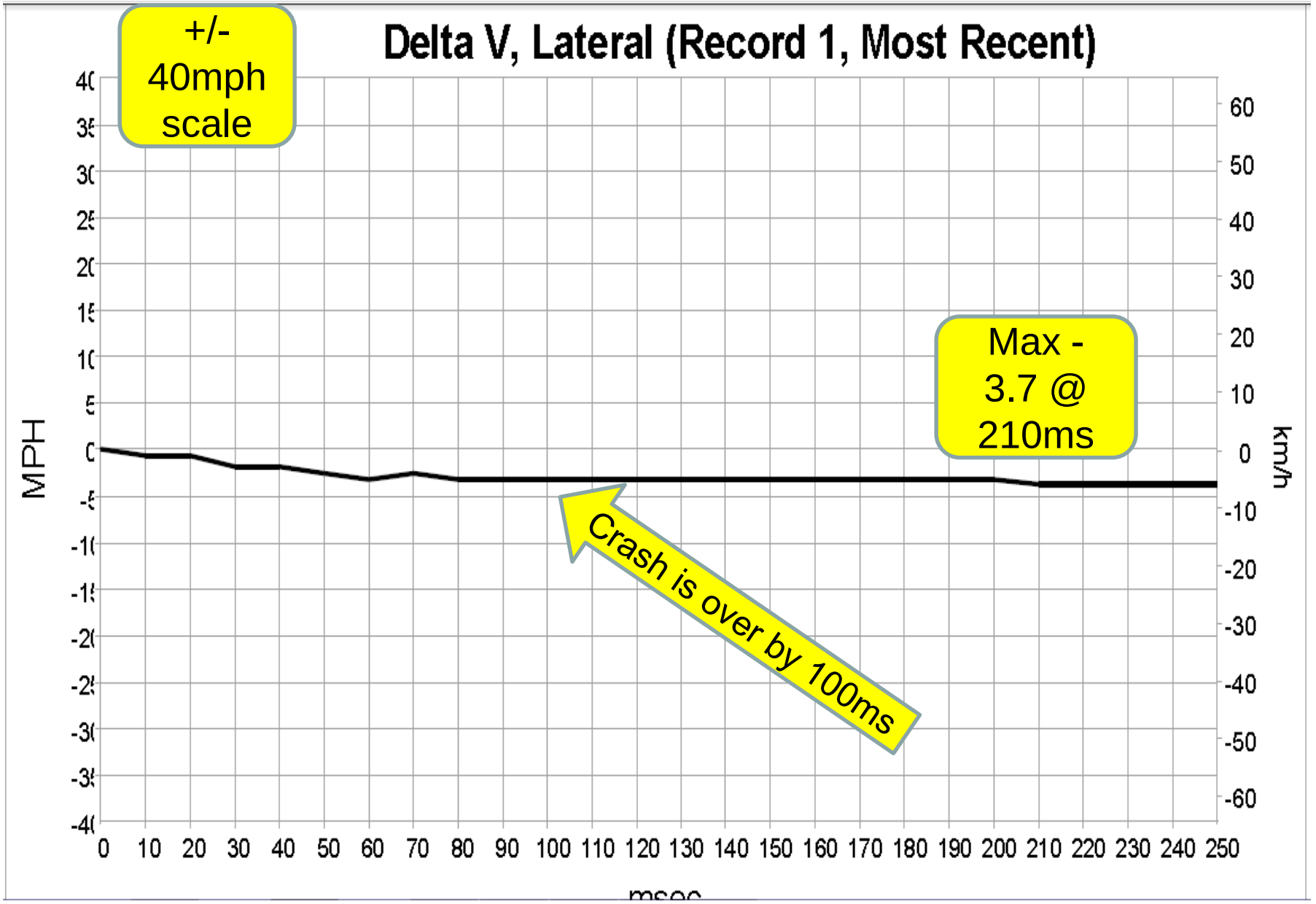
Event Type	Frontal
Time From Time Zero to Frontal Threshold (Beginning of Impact) (msec)	69
Time From Time Zero to Side Threshold (Beginning of Impact) (msec)	Threshold Not Reached
Time From Time Zero to Algo Start (Front) (msec)	1
Time From Time Zero to Algo Start (Side) (msec)	27
Time From Time Zero to Algo Start (Rear) (msec)	Algorithm Started at t0
Time From Time Zero to Deployment (Rollover) (msec)	Algorithm Not Started
Time From Time Zero to Deployment (Pitchover) (msec)	Algorithm Not Started
Time From Time Zero to Algo Start (Pedestrian Protection) (msec)	Algorithm Not Started
Maximum Delta-V, Longitudinal (MPH [km/h])	-6.2 [-10]
Maximum Delta-V, Lateral (MPH [km/h])	-3.7 [-6]
Time, Maximum Delta-V, Longitudinal (msec)	148
Time, Maximum Delta-V, Lateral (msec)	210
Clipping Time Longitudinal Sensor (msec)	Clipping Not Reached
Clipping Time Lateral Sensor (msec)	Clipping Not Reached
Multi-Event, Number of Events	1. Event
Time From Previous Event to Current Event (msec)	0
Complete File Recorded, Generic, Prio 1 Data	Completed Successfully
Ignition Cycle, Crash (cycle)	2,627
Ignition Cycle, Download (cycle)	2,732
Vehicle Mileage (km)	15,380
Operating Time (min)	47,221
Vehicle Identification Number	WDDUG8FB7EA******
Event Counter (counts)	1

Deployment Command Data (Record 1, Most Recent)

Frontal Air Bag, Time to 1st Stage Deployment, Driver (msec)	53
Frontal Air Bag, Time to 2nd Stage Deployment, Driver (msec)	253
Frontal Air Bag, Time to 3rd Stage (Vent) Deployment, Driver (msec)	Data Not Available
Frontal Air Bag, 2nd Stage Disposal, Driver	Disposal
Frontal Air Bag, 3rd Stage (Vent) Disposal, Driver	Data Not Available
Frontal Air Bag, Time to 1st Stage Deployment, Front Passenger (msec)	Data Not Available
Frontal Air Bag, Time to 2nd Stage Deployment, Front Passenger (msec)	Data Not Available
Frontal Air Bag, Time to 3rd Stage (Vent) Deployment, Front Passenger (msec)	Data Not Available
Frontal Air Bag, 2nd Stage Disposal, Front Passenger	Data Not Available
Frontal Air Bag, 3rd Stage (Vent) Disposal, Front Passenger	Data Not Available
Side Air Bag, Time to Deployment 1st Stage, Driver (msec)	Data Not Available
Side Curtain/Tube Air Bag,Time to Deployment, Driver Side (msec)	Data Not Available
Pretensioner (1), Time to Deploy, Driver (msec)	53
Side Air Bag, Time to Deployment 1st Stage, Front Passenger (msec)	Data Not Available
Side Curtain/Tube Air Bag, Time to Deployment, Passenger Side (msec)	Data Not Available
Pretensioner (1), Time to Deploy, Front Passenger (msec)	Data Not Available
Pretensioner (2), Time to Deploy, Driver (msec)	58
Pretensioner (2), Time to Deploy, Front Passenger (msec)	Data Not Available
Pretensioner (3), Time to Deploy, Driver (msec)	Data Not Available
Pretensioner (3), Time to Deploy, Front Passenger (msec)	Data Not Available

Not much use to recon – for Mercedes to know their product performed to intent


+/- 40mph scale
Delta V, Longitudinal (Record 1, Most Recent)
Max - 6.2 @ 148ms
Crash is over by 100ms
MPH
km/h
msec
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250
60 50 40 30 20 10 0 -10 -20 -30 -40 -50 -60


+/- 40mph scale
Delta V, Lateral (Record 1, Most Recent)
Max - 3.7 @ 210ms
Crash is over by 100ms
MPH
km/h
msec
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250
60 50 40 30 20 10 0 -10 -20 -30 -40 -50 -60

VW/Audi

- VW/Audi are the 13th and 14th brands to join Bosch CDR.
- VW/Audi are just phasing in EDR's starting with 2014 EOS, most models get EDR in 2015 model year - which was based on the Dec 2012 NPRM would have required EDR for Sept. 2014.

VW/Audi CDR 17.1 Coverage

See help file for Rolls Royce

Model	Location	2009-13	2014	2015	2016	2017
Last 3 digits of Bosch CDR adapter/cable numbe						
All VW/Audi Require the 387 Adapter to read DTM						
VW						
Beetle	UNDER CENTER STACK			804		
CC	UNDER CENTER STACK			804		
EOS	UNDER CENTER STACK		806			
Golf, eGolf	UNDER CENTER STACK			805		
Jetta	UNDER CENTER STACK			804		
Passat	UNDER CENTER STACK			804		
Routan	UNDER CENTER STACK	785	Chrysler Minivan Sister			
Sportwagen				not covered at this time		
Tiguan	UNDER CENTER STACK			804		
Touareg	UNDER GEAR SELECTOR			804		
AUDI				2015	2016	2017
A3,S3	UNDER CTR STACK			805		
A4, allroad	Tunnel between 1st/2nd row seats			804		500/813
A5, S5, RS5	Tunnel between 1st/2nd row seats			804		
A6, S6	Tunnel between 1st/2nd row seats			804		
A7, S7, RS7	Tunnel between 1st/2nd row seats			804		
A8, S8	Tunnel between 1st/2nd row seats			804		
Q3	UNDER CTR STACK			804		
Q5, SQ5	Tunnel between 1st/2nd row seats			804		
Q7	Under Center Stack					500/813
R8	Under Center Stack					804

No “back model” coverage

Other VW Brands & Info

- No word on Porsche yet- but it is only 0.3% of the US market
- 4 Bentley models 2016+ now covered - 0.0%
- Lamborghini Adventadore & Hurracan covered 2015+

VW/Audi Details

- Same data set for all event types
- Time from last precrash to AE noted
- Speed, brake, accel pedal sampled at same time (not asynchronus like GM)
- Recorder holds up to 6 ND events in Chronological order.
- New modules will use Flex Ray
- Date/Time of Crash from clock (GPS if clock not set)

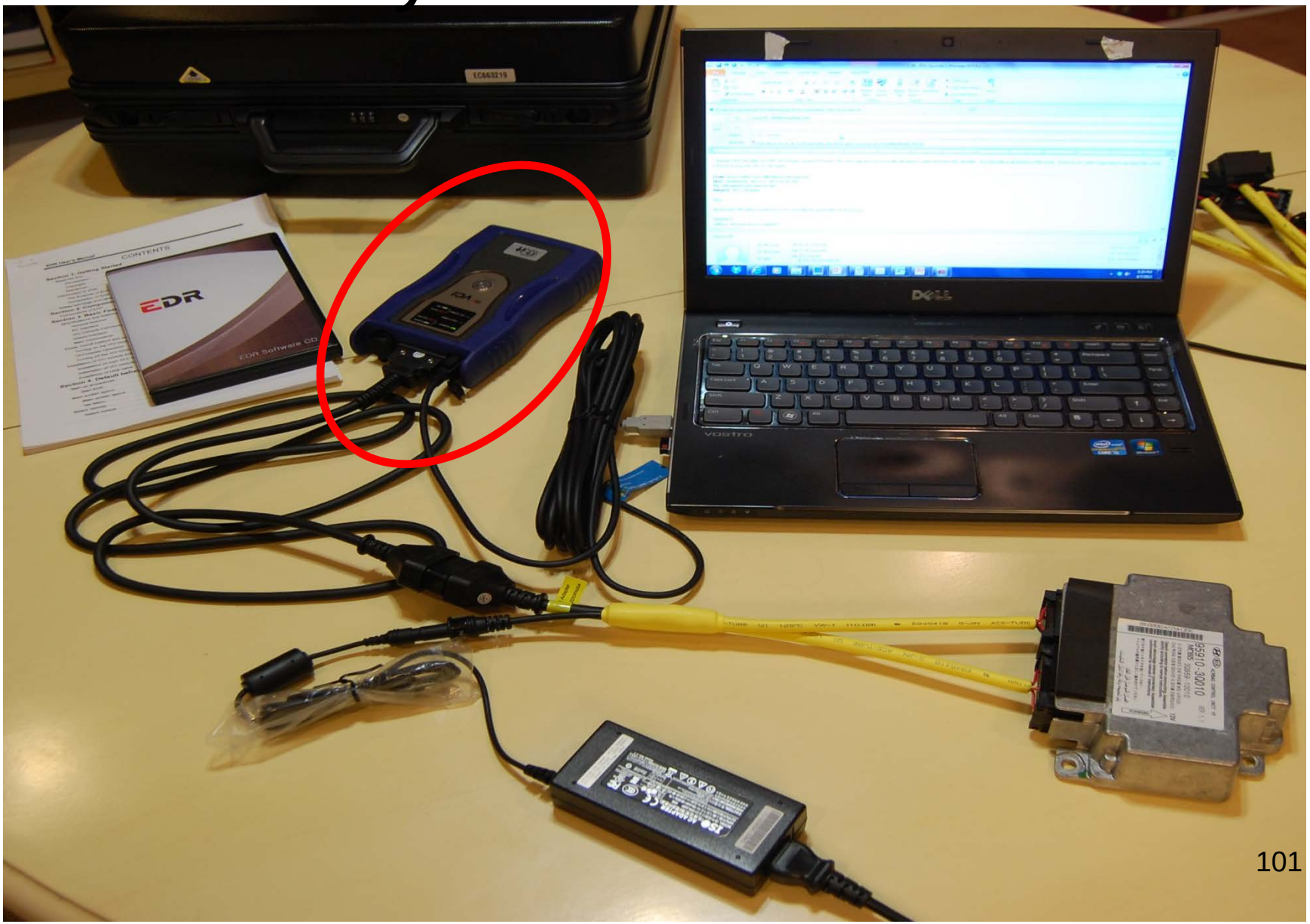
VW/Audi Details

- Modules live for 15 sec after power out
- Image time 1-5 minutes
- Pushbutton ignition – only 1 push to ACC
- Battery disconnect after serious crashes may force backpowering
- Must power ACM **and** gateway module

Hyundai/Kia Tools

- Hyundai: Call GIT America Tool Company (714)433-2180 order Part G0Z**H**DMN001 ***$4700***
- Kia" Call Dealer Tools hotline 888-542-1011, order part GIT0Z**K**DMN001 ***$6486 + tax+ship***
  www.kiaspecialtools.com
- Software updates $295/yr from GIT Tool Company
- K/H Interface modules are identical, but you must buy both to get software for both. Software updates won't download unless the correct interface module is attached to your laptop.
- I read modules for others and will rent my kit(s) out for $500

The Hyundai EDR Reader



Hyundai/Kia Tool Info (not new)

- The tool “officially” only reads 2013 and later models – there is a menu to pick model year and 2013+ is the only choice.
- But it communicates with several 2010-2012 models.
- However, your report will say you told the software it is a 2013.
- Admissibility could be a concern. Data successfully admitted in Virginia and California.

Kia/Hyundai GIT Tool Oct 2016

Model	Vehicle Size, Descrip	Model Years 2010	2011	2012	2013*	2014	2015	2016	2017
KIA		Cable #	Cable #	Cable #	Cable #	Cable #	Cable #	Cable #	Cable #
CADENZA (VG, 2014) (Azera sister)	112.0" WB	YG requires cable 13 per software				19(VG)	19	19(VG)	19
FORTE (TD 2013, YD 2014))	104.3"WB, compact	10	10	10	10 (TD)	6 (YD)	6	6	6
K900 (KH 2015)	119.9"WB						11 (KH)	11	11
OPTIMA (2013 TF QF, 2016 JF, JFA)	110.0" WB sedan	No	4	4	4 (TF/QF)	4	4	14 (JF,JFA)*	14
OPTIMA (TF HEV, JFHEV)	110.0" WB sedan	No	4	4	(TF HEV)	4	4	4	23(JFHEV)
RIO (UB)	94.5"WB, compact	No	No	8	8(UB)	8	8	8	8
RONDO (Canada only 2011 MY+)		No	out of production						
SEDONA (VQ '14, YP '15)	113.8/118.9" WB miniv	No	No	No	No	10(VQ)	12(YP)	12	12
SORENTO (XM to 2015)(UMA 2016)	106.3" WB, crossover	No	3	3	3(XM)	7(XM)	7	13 (UMA)	13
SOUL (AM 2013, PS 2014+)	100.4" WB	1 see note 1	1 see note 1	1	1(AM)	12(PS)	12(PSEV)	12	12
SPORTAGE (SL)	103.9" WB, crossover	No	Yes	2	2(SL)	2	2	2	13(QL)
HYUNDAI									
ACCENT (RB)	94.5"WB, compact	No 10	No?	5 see note 2	5(RB)	5	5	5	5
AZERA (HG)	109.4"WB, sedan		likely	Yes	7(HG)	7	7	7	7
ELANTRA Coupe(JK), Sedan(UD/MD)	104.3"WB, compact	No	Yes	8	8(JK/UD/MD)	8	8	8	21(AD, ADA)
ELANTRA GT (GD) (aka "i30")	former Elantra Touring	old model		new model	6(GD)	6	12	12	12
EQUUS (VI)(stretch genesis)	119.9" WB, luxury sed	?	yes?	yes?	9(VI)	11(VI)	11	11	
GENESIS (Coupe BK, Sedan BH/DH))	115.6" WB, luxury sed	9 shave keyway	9 shave keyway	?	9(BK, BH)	9	13(DH)	13	13
GENESIS G90									13(HI)
SANTE FE (Sport DMA, NC)	106.3" WB, crossover	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)	7	7	7	7
SONATA (YF sedan, YF HEV)	110.0" WB sedan		4	?	4(YF)	4	14 (LF) 4 (HEV)	14 (LF HEV)	14
TUCSON (LM to 2015)(TL in 2016)	103.9" WB, crossover	Yes	Yes	2	2(LM)	2	2	13(TL)	13
VELOSTER coupe (FS)	104.3"WB, compact	out of productio		?	5(FS)	5	5	5	5

Kia/Hyundai Tool Issues

- The tool originally did not ask for a VIN, (new software does), only model and model year. VIN not stored in the ACM. Do sufficient documentation to prove your readout came from your vehicle.
- The K/H report is a secure PDF. There is no raw data file. If K/H ever discovers an error and updates software, you will have to re-read the module. The most recent release now updates software version in the report
- Recommended: ***Secure module*** as evidence

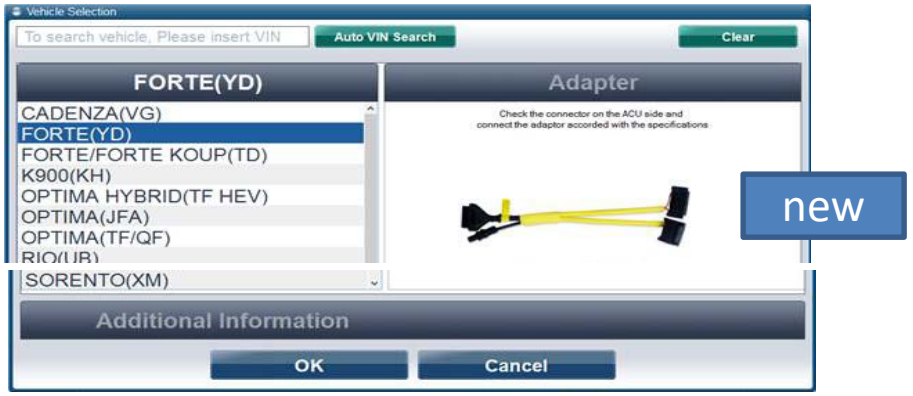
K/H 2010-2012 data reliability?

- Internal sources say SOME data elements on the 2010-2012's may not be correct.
- In 50+ NHTSA crashes, speed, delta V and belt use appear OK
- Steering angle, and accel data appear implausible on some
- Key cycles at download not in all models
- SAE 2014-01-0502 has details. Additional testing on 2012 Soul reported in SAE 2015-01-1445 @ SAE Congress.

New Software Kia V11, Hyundai V13

- New look and feel to screens – eliminates MY/engine, requires VIN entry instead.
- The VIN entry is “hidden” behind the display of the cable needed for direct to module download, you must click on the cable picture





- Auto VIN detect works on 2013+. A surrogate 2013 VIN will be required for 2010-2012 models. (now at Kia V15, Hyundai V17)

New Report Header Format 4/2016
Kia V11, Hyundai V13

EDR

Vehicle Information

KIA | SPORTAGE(SL) | 2014 | AIRBAG SYSTEM	
VIN as Programmed into EMS	

Additional Information

User-entered VIN	KNDPCCA61E7669127
User Name	Rich Ruth Test
Case Number	687165487
Crash Date	033116
Saved-on Date	2016-03-31 14:47
EDR Tool Version	E-N-K-01-00-0011
EDR Report Version	EDR001-R01
Tire Size(s)	245/60/17
Memo	Test

New better Data limitations

- Data Limitation

General Information:

Tools for downloading and interpreting the EDRs in Kia vehicles have been developed for vehicles produced after September 1, 2012. Currently, there is no tool for downloading and accurate interpreting data from the EDRs in Kia vehicles produced prior to this date.

EDR(Event Data Recorder):

- The EDR function is part of the Airbag Control Unit(ACU).

- ACU can store up to two events.

- Event means a crash or other physical occurrence that causes the trigger threshold to be met or exceeded, or any non-reversible deployable restraint to be deployed, whichever occurs first:

  1. Deployment Event:
     1) the event which is recorded if an airbag is commanded to deploy.
     2) the event is locked and cannot be overwritten.

  2. Non-deployment Event:
     1) the event which is recorded, but in which an airbag is not commanded to deploy
     2) the event is not locked and can be overwritten by a subsequent event (Deployment or Non-deployment event), for example, Pretensioner(s) only deployment
     3) An example of a non-deployment event is a pretensioner-only deployment with no airbag deployments

- Ignition cycle count will increment by 1 in the following cases

  1. the power mode change from OFF/Accessary to IGN ON/RUN

  2. EDR data download by tools

- The ACU can record data for all or some of the following events. But, depending on the vehicle's configurations, data for side crash and/or rollover crash(event) may not be recorded.

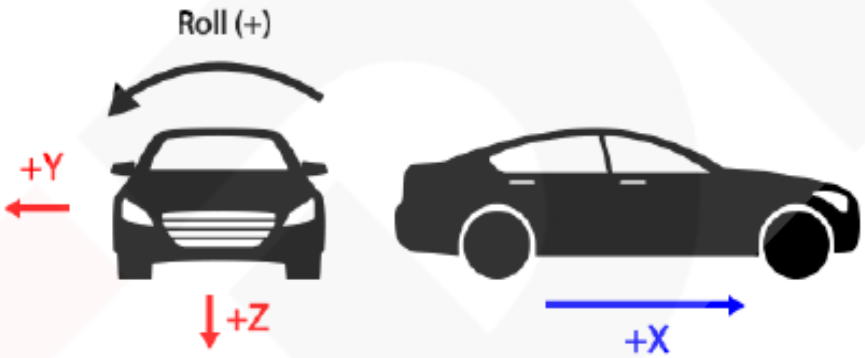
Data Limitation New better Data limitations

Data Element Sign Convention:

The following table provides an explanation of the sign notation for data elements that may be included in the EDR report. Directional references to sign convention are from the point of view of the driver.

Data element name	Positive sign	Note
Longitudinal acceleration	Forward direction	+X at the figure 1
Delta V, longitudinal	Forward direction	+X at the figure 1
Lateral acceleration	Left to Right direction	+Y at the figure 1
Delta V, lateral	Left to Right direction	+Y at the figure 1
Normal(vertical) acceleration	Downward direction	+Z at the figure 1
Vehicle roll angle	Clockwise about the longitudinal axis	Roll(+) at the figure 1
Steering input	Counterclockwise rotation	-

Figure 1. Sign Conventions



New Summary Page

EDR Information

Part No. (EOL Code) as programmed into ACU	95910-3W000(SL90)
ECU SW Version as programmed into ACU	28304666
EDR Version as programmed into ACU	

< Event 1 >

Event Status at Event

Multi-event, Number of Event (1 or 2)	1 event
Time from Event 1 to 2 [msec]	0
Completed File Recorded (Yes or No)	YES
Ignition cycle, crash [cycle]	56
Ignition cycle, download [cycle]	Invalid data

Pre-Crash Information (-5 ~ 0 sec)

Time [sec]	Vehicle Speed [kph]	Engine RPM [rpm]	Engine Throttle [%]	Service Brake [on/off]	ABS Activity [on/off]	Stability Control [on/off/engaged]	Steering Input [degree]
-5.0	0	0	4	OFF	OFF	on	0
-4.5	0	0	4	OFF	OFF	on	0
-4.0	0	0	4	OFF	OFF	on	0
-3.5	0	0	4	OFF	OFF	on	0
-3.0	0	0	4	OFF	OFF	on	0
-2.5	0	0	4	OFF	OFF	on	0
-2.0	0	0	4	OFF	OFF	on	0
-1.5	0	0	4	OFF	OFF	on	0
-1.0	0	0	4	OFF	OFF	on	0
-0.5	0	0	4	OFF	OFF	on	0
0.0	0	0	4	OFF	OFF	on	0

Close up of new data table

Pre-Crash Information (-5 ~ 0 sec)

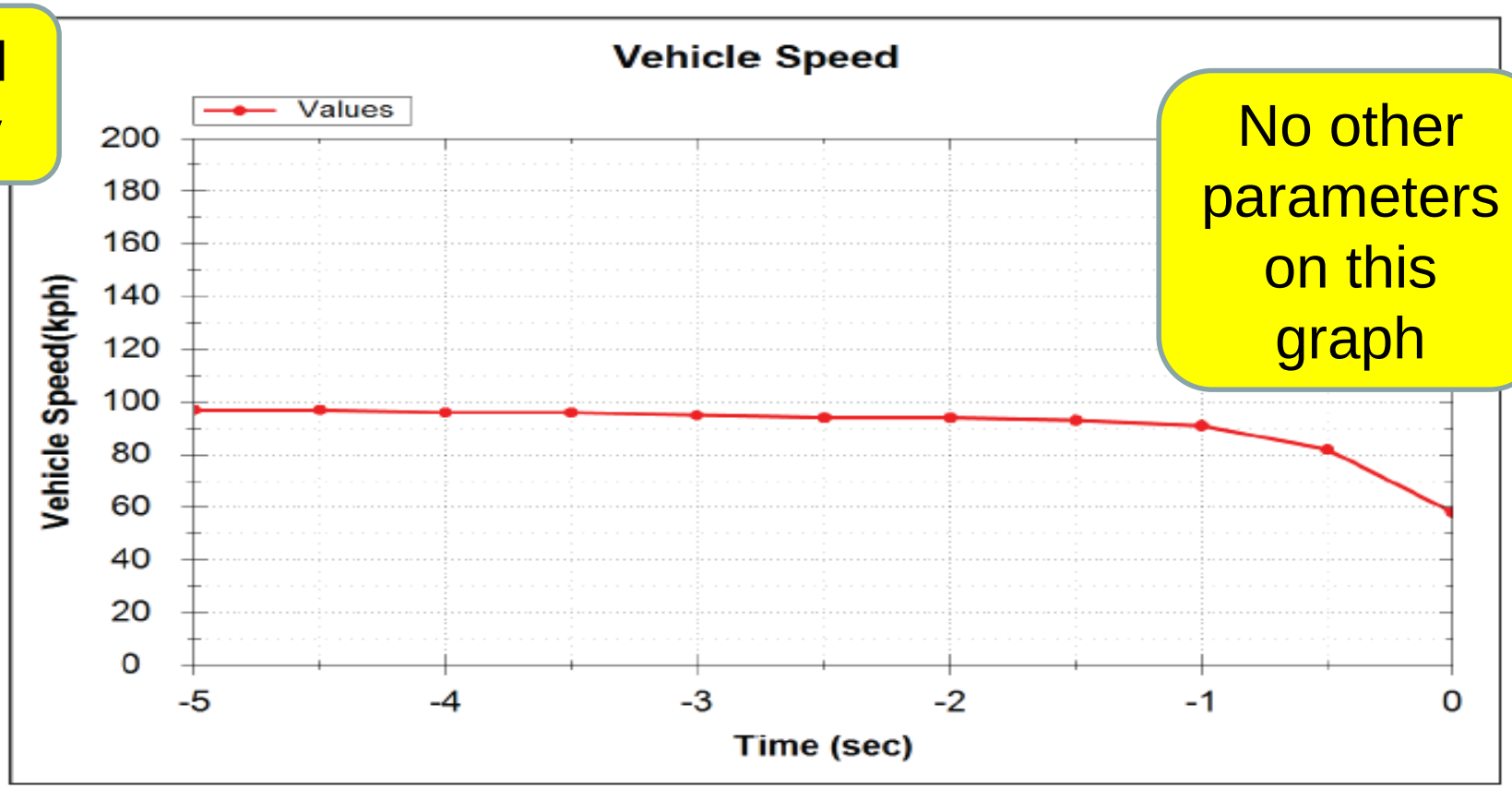
Time (sec)	Vehicle Speed [kph]	Engine RPM [rpm]	Engine Throttle [%]	Service Brake [on/off]	ABS Activity [on/off]	Stability Control [on/off/engaged]	Steering Input [degree]
-5.0	0	0	4	OFF	OFF	ON	0
-4.5	0	0	4	OFF	OFF	ON	0
-4.0	0	0	4	OFF	OFF	ON	0
-3.5	0	0	4	OFF	OFF	ON	0
-3.0	0	0	4	OFF	OFF	ON	0
-2.5	0	0	4	OFF	OFF	ON	0
-2.0	0	0	4	OFF	OFF	ON	0
-1.5	0	0	4	OFF	OFF	ON	0
-1.0	0	0	4	OFF	OFF	ON	0
-0.5	0	0	4	OFF	OFF	ON	0
0.0	0	0	4	OFF	OFF	ON	0

Event 1-2 (second section of memory)

After summary, one data element at a time

1 [Vehicle Speed]

KPH only



No other parameters on this graph

Num	Time (sec)	Vehicle Speed (kph)
1	-5.0	97
2	-4.5	97
3	-4.0	96

Other Non-Bosch CDR Tools

Law Enforcement: You can't afford this, ask the dealer to help you

	2012 Nov YTD SALES	Nov YTD US Market SHARE	2013 EDR?	Tool?
SUBARU	299,788	2.3%	YES	Hitachi HDS-3000
Dealer scan tool $2835 plus software $1835/mfr (905)669-4812				
New Denso tool from Nuspire required for 2016+, $2K + $5800 software/yr				
MITSUBISHI	53,677	0.4%	YES	OTC tool co
Panasonic Touchpad + software $6395, annual $1200/yr, 888 727 6672				
JAG/LAND ROVER (Tata Motors India)	49,452	0.4%	YES	OTC/Bosch SPX
Full dealer scan tool, Toughbook & software approx $5000				
SAAB	480	0.0%	YES?	Hitachi HDS-3000?
Same hardware as Subaru, software $1835 for Saab				

SUBARU

- Subaru is believed to have released an EDR to 563 regulation intent in the all new 2012 Subaru Impreza. As Subaru introduced new models, they added EDR.
- Subaru met the publicly available tool requirement by adding the EDR readout function to its dealership scan tool.
- The airbag system diagnosis menu has a “read EDR” tab within it.

Subaru Tool Coverage (not CDR)

Model	2011	2012	2013	2014	2015	2016	2017
SUBARU		<-------Old Hitachi tool or new Denso Tool----->				New Denso Tool	
BRZ	n/a	n/a	Yes (Scion FRS sister)	Yes	Yes	Yes	Yes
Impreza	No	Yes	Yes	Yes	Yes	Yes	Yes
WRX/STI	No	No	No	No	Yes	Yes	Yes
Legacy	No	No	Yes	Yes	Yes	Yes	Yes
Outback	No	No	Yes	Yes	Yes	Yes	Yes
Forester	No	No	No	Yes	Yes	Yes	Yes
Tribeca	No	No	No	No	Discontinued		
Crosstrek	n/a	n/a	Yes	Yes	Yes	Yes	Yes

NOTE: The Subaru tool is a dealership fault code tester made to work in thru the DLC in vehicles with functioning electrical systems only.

Instructors may be able to fabricate a direct to module cable from a vehicle side connector

Subaru EDR Tool

- Subaru used the Blue Streak Electronics “Subaru Diagnostic Interface” hardware with “Subaru Select Monitor III” software from its first EDR’s in 2012 thru 2015. This is approximately $3K for hardware and $2K for software annual subscription.
- For 2016 Subaru has gone to a new Denso DSTi diagnostic interface with new software from Nuspire. Hardware $2K software $5.8K/yr, is compatible to ‘04.

Subaru

- The Dealer tool is DLC only. The electrical system must be working, or you must backpower both ACM and DLC fuses
- Or put ACM in surrogate vehicle.
- Direct to module cables can be fabricated.
- Tutorial available on software use
- Breaking news – 2015 tool may still work on 2016 and 2017 models!

Subaru data labeling issues

Column Headings generated as found in CSV file

Sampling time	Event name	Vehicle Speed	Accelerator Opening Angle %	Service brake	on/off	Engine Speed	Motor RPM	ABS activity	Stability control	Steering input	Steering input (%)	Mark
sec		km/h	%		rpm	rpm			deg	%		

Corrected Column Headings based on other data views available in software

		Vehicle Speed km/h	Accelerator Opening Angle %	Service brake on/off	Engine Speed rpm	Motor RPM rpm	ABS Activity	Stability Control	Steering input degrees	Steering input %	Mark?	Suggested time label
0	Frontal Crash New Data	21	0	OFF	1300	invalid	OFF	ON	0	0	0	-5
0.5	Frontal Crash New Data	21	0	OFF	1200	invalid	OFF	ON	0	0	0	-4.5
1	Frontal Crash New Data	20	5	OFF	1100	invalid	OFF	ON	0	0	0	-4
1.5	Frontal Crash New Data	21	0	OFF	1500	invalid	OFF	ON	0	0	0	-3.5
2	Frontal Crash New Data	21	0	OFF	1300	invalid	OFF	ON	0	0	0	-3
2.5	Frontal Crash New Data	21	0	ON	1000	invalid	OFF	ON	0	0	0	-2.5
3	Frontal Crash New Data	20	0	ON	1100	invalid	OFF	ON	0	0	0	-2
3.5	Frontal Crash New Data	18	0	ON	1100	invalid	OFF	ON	-2.5	0	0	-1.5
4	Frontal Crash New Data	17	0	OFF	1100	invalid	OFF	ON	-2.5	0	0	-1
4.5	Frontal Crash New Data	16	0	ON	1100	invalid	OFF	ON	-7.5	-1	0	-0.5
5	Frontal Crash New Data	11	0	ON	900	invalid	OFF	ON	-2.5	0	0	0

There are no data limitations given

MITSUBISHI 0.5% market share

- Mitsubishi originally released an EDR tool based on their dealership tool that included a dedicated Panasonic Toughbook Laptop and cost $9K.
- They recently changed the kit to work with a Panasonic Touchpad. Cost reduced to $6380.
- The kit is made to work only on vehicles with a functioning electrical system thru the DLC.
- Info to date indicates Mitsubishi has a gateway module that will require duplicating gateway in cable to make it work DTM.

Mitsubishi EDR Tool (NOT CDR)

Model	2008	2009	2010	2011	2012	2013	2014	2015	2016
MITSUBISHI									
Eclipse		?	Proprietary EDR Tool			Not in production			
Eclipse Spyder		?	Data in deployments only?						
Endeavor		?							
Galant		?							
iMiEV					Public EDR Tool				
Lancer		2.3 sec precrash @ 0.1 sec intervals				Public EDR tool 5 sec @ 0.5 sec			
Lancer Evolution		?	Law enforcement call Mitsu			Part 563 compliant			
Lancer Sportback		?							
Mirage (subcompact)		Not in production			Equipped?	?	?	?	?
Outlander		?		Public EDR Tool					
Outlander Sport					Public?				

Jaguar/Land Rover

- Jag/LR have EDR's, they are read with a dedicated laptop and software.
- The tool only retrieves the hex data, the hex must be sent to England for interp.
- Jag/LR have now designated 2 engineers in New Jersey HQ as EDR contacts.
- **They are now referring requesters to <u>suppliers.</u>** (call to find out who supplier is)
- Some Mid 2000's have ***PCM data*** (when Jag/LR was owned by Ford) but *not spoofable* with CDR.

Newly published research

- Every year at the SAE World Congress in April new research is presented.
- A very brief summary follows, I encourage you to read these papers.
- Other good sources of EDR info include Collision Magazine and Accident Reconstruction Journal.

2016 New EDR Research

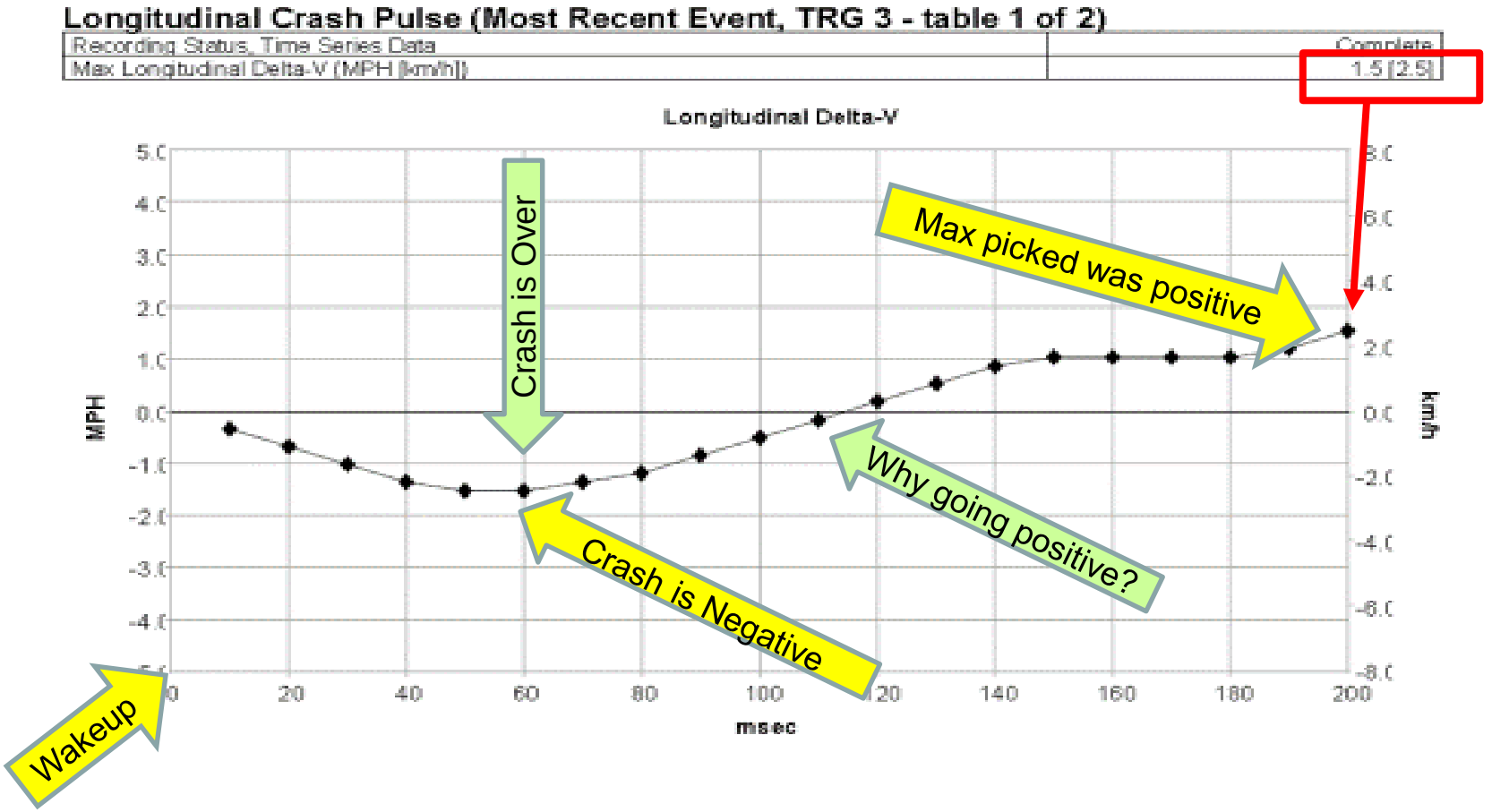
SAE 2016-01-1494 Comparison of Accuracy and Sensitivity of Gen 1,2 and 3 Toyota EDR's in Low Speed Collisions, Xing et al, MEA

SAE 2016-01-1495 EDR Developed by Toyota, Iyoda et al, Toyota

SAE 2016-01-1496 Longitudinal Delta V offset between Front and Rear Crashes in a 2007 Yaris, Ruth and Muir

SAE 2016-01-1497 "Compendium of Pass Vehicle EDR Literature and Analysis of Validation Studies" Bortles et al

Toyota Gen 1&2 offset in Delta V



Under reports frontals, over-reports rears

EDR Regulatory Update

- Part 563 standardized minimum content for vehicles voluntarily equipped with EDR, became effective Sept 2012.
- NHTSA required 10% accuracy on Delta V Later clarified, 10% within range of sensor, requires OEM to report WHEN sensor first exceeds range (clips), effective 9/1/2014.
- Accuracy can be worse than +/-10% after clipping occurs

Regulatory Update

- NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (past!)
- NPRM also proposed EDR be made an FMVSS 405 (Safety Standard), OEM's would have to self certify EDR's. Strong OEM objections were filed.
- Still "Pending" as of this printing, once published followed by 60 day comment period before becoming effective **(date keeps slipping every time I check)**.

Regulatory Update

- Senate Bill 1925 “Driver Privacy Act” was introduced 2014 with bi-partisan support, to require warrant or owner consent prior to accessing EDR. Also directs NHTSA to study what data is needed. Passed as part of highway funding bill, signed into law Dec. 4, 2015 by President Obama.
- Requires NHTSA to study how much data is really needed & report to congress

Regulatory Update

- NHTSA/IIHS got 10 auto mfrs to agree on 9/11/2015 to making Auto Emergency Braking standard (timeline TBD). Voluntary commitment is a relatively new strategy by NHTSA.
- Next big issue is Autonomous Vehicles (self driving cars).
- States are drafting conflicting regulations on type of certification required – could hinder development.

EDR admissibility update

- “Every year EDR evidence is admitted at the trial level in hundreds of cases and we never hear about them.” - Prosecutor Bill Melkonian at the 2014 CDR User’s Conference Legal Update. No change.
- Warrants now required, admissibility shifts to Probable Cause for getting that warrant. For law enforcement, it’s also about the training and experience of the officer applying for the warrant

Not reading EDR an Issue?

- People v Ortega – 2012, CA (nonpublished/noncitable) – EDR not read, defendant claimed it would have exonerated him.
- Court rules “duty to preserve evidence exists only when 1) the exculpatory value was apparent before the evidence was destroyed and (2) the defendant could not obtain comparable evidence by other, reasonably available means.

Virgin Islands v Milligan (2013) –

- Defendant claims he was deprived of due process because police did not read EDR and car was given back to insurance company and no longer available to read.
- Court ruled no bad faith, denied motion to dismiss. Determined evidence was only “potentially” useful.

Emerging Trends

- Kia/Hyundai Pre-2013 data
  - 2012 Kia Soul data admitted in CA criminal case – Vandiver literally ran a car crash to validate the data.
  - 2011 Hyundai Sonata admitted in VA criminal trial 4/7/2016 (Ruth)
- Unsupported Ford PCM data
  - “VIN spoofed” data passed Frye hearing in Colorado Criminal case March 2016 (Ruth)
- Mitsubishi incomplete file recorded admitted in CA (Vandiver)

Civil Cases

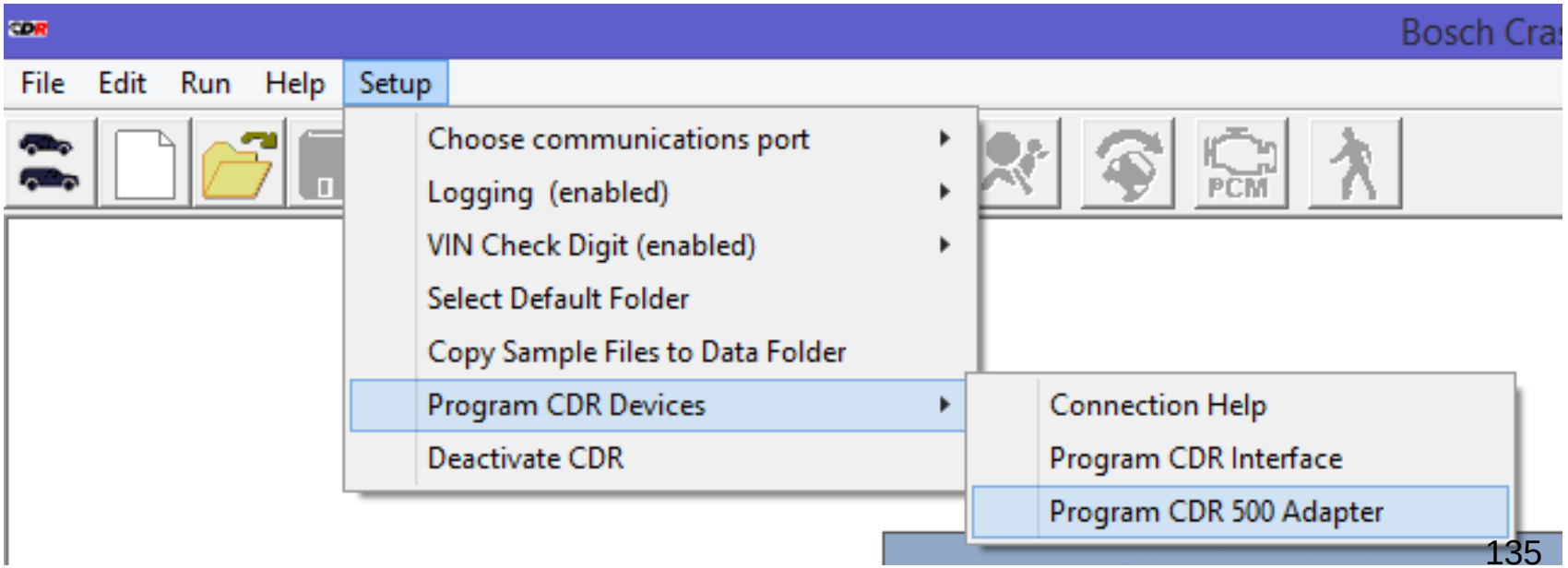
- Niemeyer V Ford (2012) plaintiff lost vehicle, court found no bad faith but that Ford was prejudiced by not having car to inspect.
- Plaintiff raised “source code not available” issue. Bosch employee called as witness.
- Lewis V New Prime, Inc – (2013, GA) - Court ruled defense expert could testify about speed derived from EDR Delta V – that accuracy could be addressed via cross exam. Plaintiff claimed defendants authorized moving truck from crash scene resulting in destruction of evidence. Jury verdict for defendants.

Bosch CDR Hardware Update

- *No **functional** change* to **interface** module since ‘05 -Blue CAN Plus, Green, and new Green “CE” (like UL) are all OK
- Bosch has managed new technology with adapters so as not to have to make everyone update the interface module regularly.
- Electric Vehicles are exempt from OBD-II requirements, may not support VIN requests.

(Relatively) New Adapters

- **CDR 500 “Flex Ray” direct to module adapter for BMW hybrids & some VW/Audi, more to come, cost $2500**
- **ACM only communicates in flex ray, the CDR 500 translates from vehicle CAN to Flex Ray.**
- **Flex Ray module requires firmware updates now with newer software (see below)**



More Hardware Updates

- Proliferation of **DLC adapters** as manufacturers use more than 1 can bus (low speed, medium speed, high speed) and Bosch must connect to nonstandard pins in the DLC to find the correct CAN bus for the ACM
- New ACM's have big capacitors with big inrush currents to charge them. Must use #2435 2.5 amp current limited power supply 387 adapter or risk damaging it

HARDWARE CHECKLIST 17.1

1699200127	Cable, ID811 CDR Mazda ACM	16.1
1699200128	Cable, ID812 CDR Mazda ACM	16.1
1699200126	Cable, ID810 CDR Honda ACM	16.2
1699200261	Cable, ID816 CDR Volvo ACM	16.2
1699200264	Cable, ID819 CDR FCA ACM	16.2
1699200369	Adapter, DLC 3	16.2
1699200129	Cable: Audi ACM (Cable ID#813, requires CDR 500 Adapter)	16.3
1699200143	Cable, ID814 CDR Toyota ACM	16.3
1699200144	Cable, ID815 CDR Toyota ACM	16.3
1699200262	Cable, ID817 CDR Toyota ACM	16.3
1699200263	Cable, ID818 CDR Toyota ACM	16.3
1699200365	Cable, ID821 CDR GM ACM	16.3
1699200953	Cable, USB to Serial, 6', DB9 Male	16.4
1699200428	Cable, ID823 CDR Ford ACM	16.5

New 3-digit Part ID, not necessarily last 3 digits of longer official Bosch Part Number

2015: Toyota Cables for non US markets released. Not required in US.

New Cable 824 for Karma	17.0
New Cable 822 for BMW	17.0

New License Holder Info 17.0

CDR File Information

User Entered VIN	1G3NL52T31C262502
User	Jesse Kirk
Case Number	Online EDR Technician
EDR Data Imaging Date	09/19/2016
Crash Date	09/19/2016
Filename	9287_1G3NL52T31C262502_ACM.CDRX
Saved on	Monday, September 19 2016 at 14:29:37
Imaged with CDR version	Crash Data Retrieval Tool 17.0
Imaged with Software Licensed to (Company Name)	Collision Reconstruction Associates
Reported with CDR version	Crash Data Retrieval Tool 17.0
Reported with Software Licensed to (Company Name)	Ruth Consulting LLC
EDR Device Type	Airbag Control Module
Event(s) recovered	Deployment

Recent CDR Software Updates

- 16.0 March 8, 2015 – VW, Lamb, Bentley
- 16.1 May 25, 2015 – some 2016's
- 16.2 August 10, 2015 – more 2016's
- 16.3 Nov. 20, 2015 – Toyota worldwide!
- 16.4 Dec. 17, 2015 – GM 2017, Ford Korea
- 16.5 Mar. 18, 2016- GM/Ford/Honda 2017's
- 16.6 May 23, 2016 – FCA/Volvo/VW 2017's
- 17.0 August 26, 2016 – License holder ID, more 2017's, new Karma brand, Korea/China
- 17.1 Nov. 2, 2016 – more 2017 models

The 17.1 Story

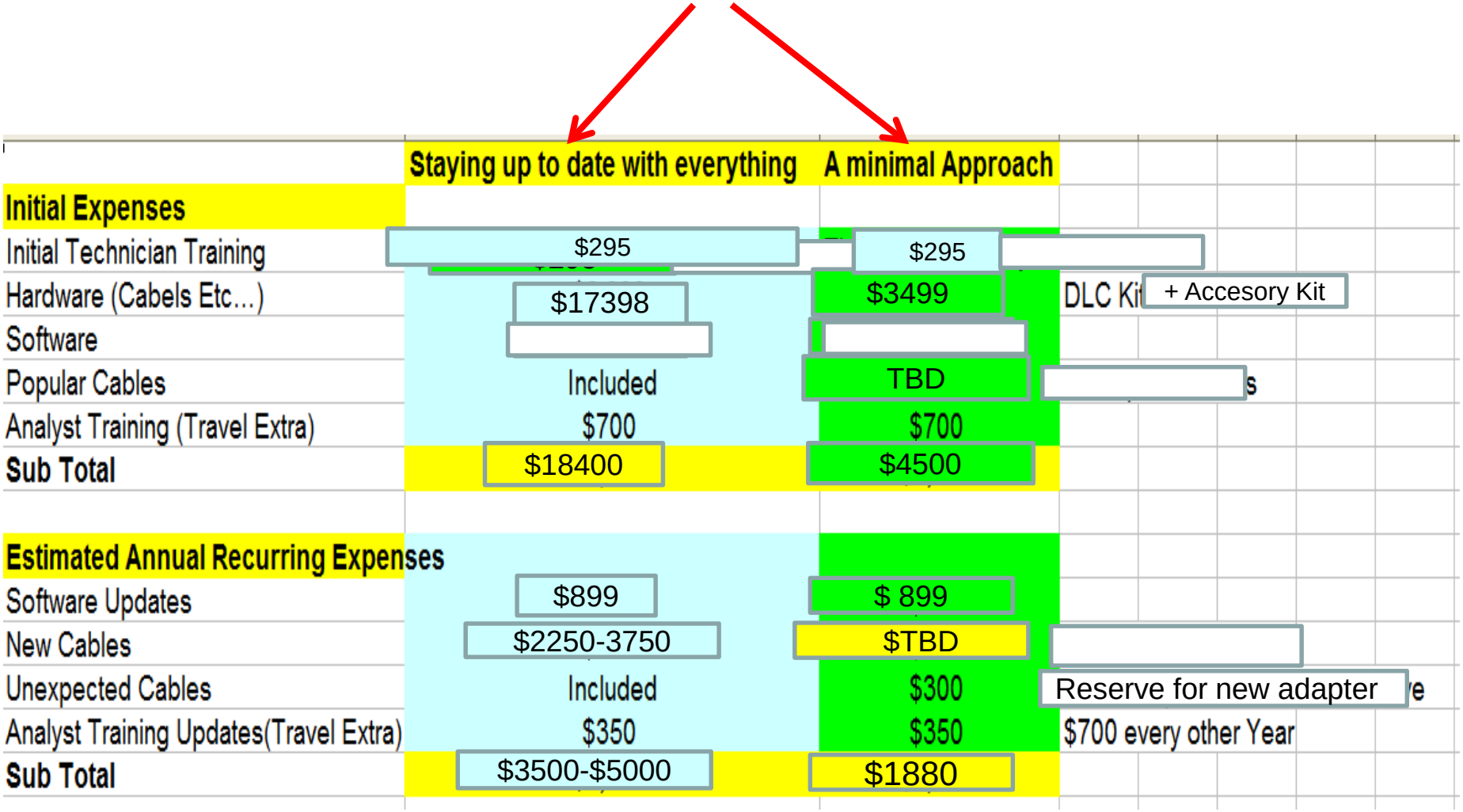
- Every year we get a new model year of coverage, and every year another year old cars without EDR's get off the road. The covered vehicles on the road with CDR is 50% and climbing. The total available from manufacturers and other tools is much more!!
- We are SO LUCKY to be in the USA with an EDR regulation. Europe, Australia, Korea, and China are just getting started!

17.0 reverse compatible but 16.6 is not “forward compatible”

- 17.0 has a new file type to accommodate the new licence holder information
- 17.0 will open all older file types
- 16.6 cannot open files saved in 17.0. You will get an error message saying “this is not a valid CDR file”.
- If you need to view a 17.0 file and don’t have 17.0 software, you can always download the free reader version.

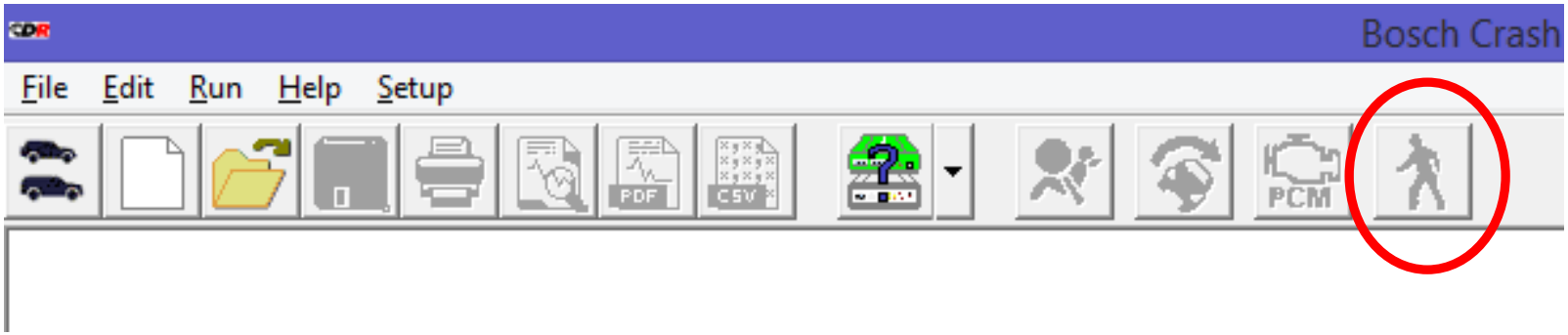
CDR Budgeting

Two approaches to a budget

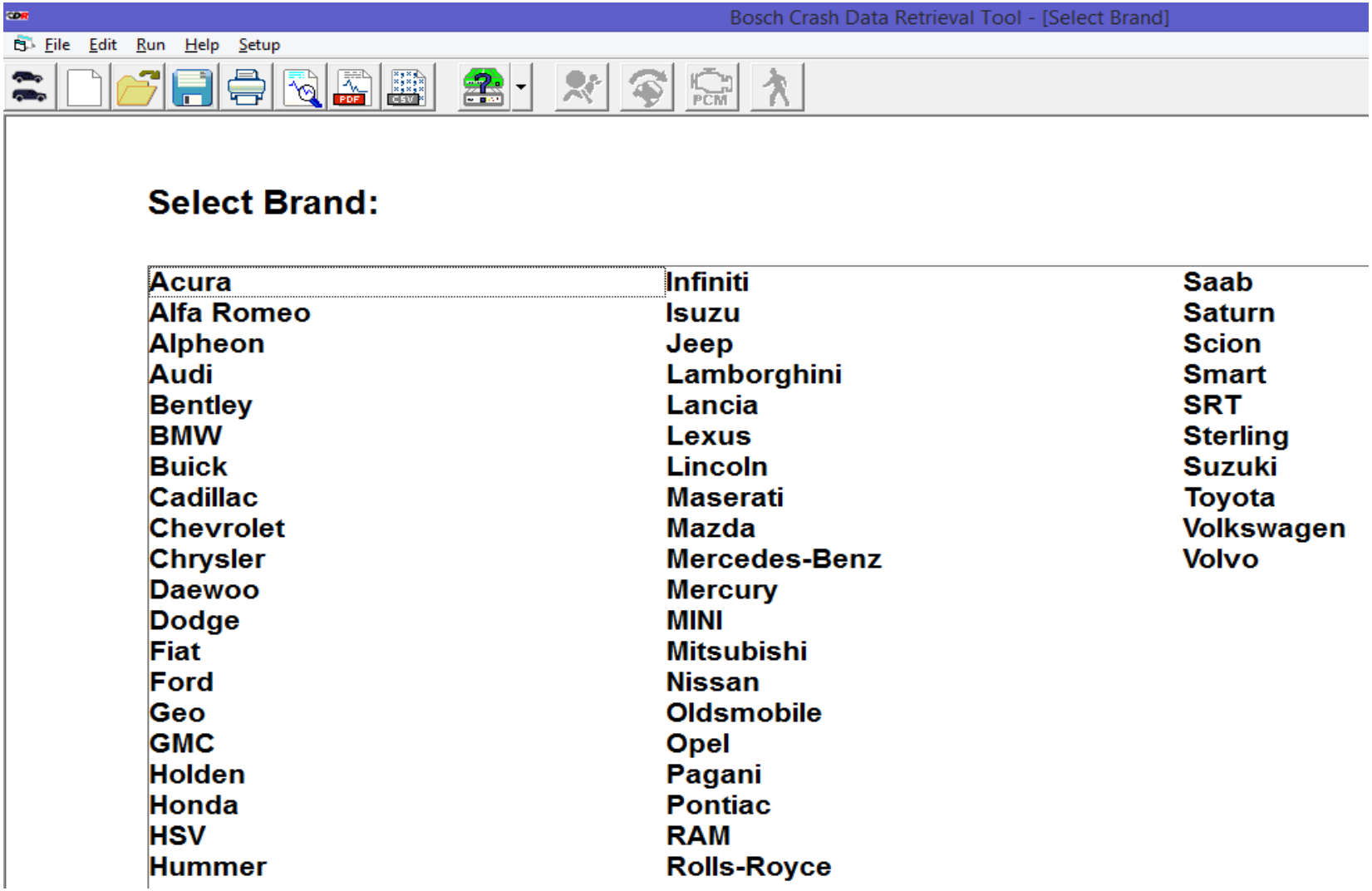


	Staying up to date with everything	A minimal Approach	
Initial Expenses			
Initial Technician Training	$295	$295	
Hardware (Cabels Etc...)	$17398	$3499	DLC Kit + Accesory Kit
Software			
Popular Cables	Included	TBD	s
Analyst Training (Travel Extra)	$700	$700	
Sub Total	$18400	$4500	
Estimated Annual Recurring Expenses			
Software Updates	$899	$ 899	
New Cables	$2250-3750	$TBD	
Unexpected Cables	Included	$300	Reserve for new adapter e
Analyst Training Updates(Travel Extra)	$350	$350	$700 every other Year
Sub Total	$3500-$5000	$1880	

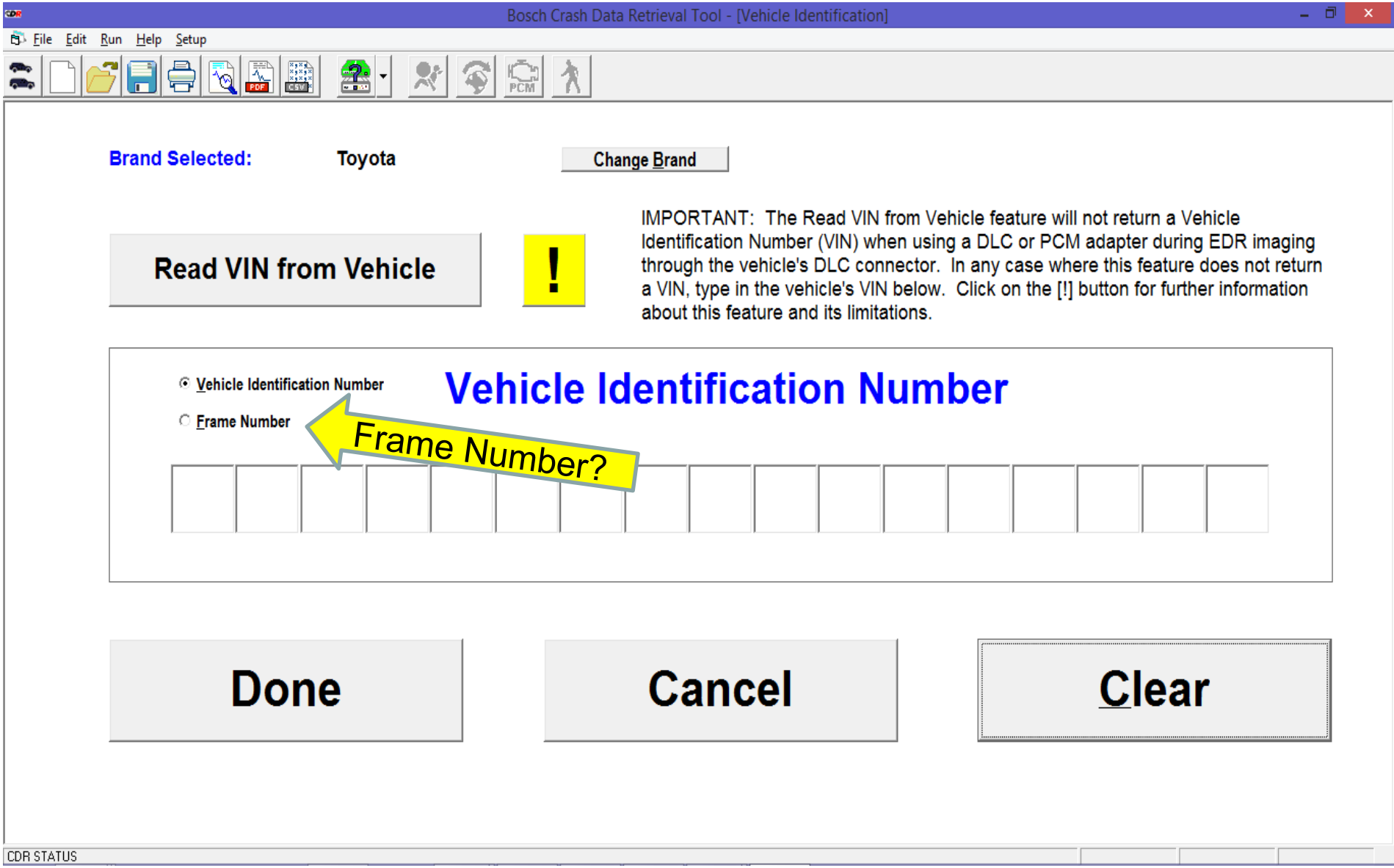
New Pedestrian Protection Module with Data



CDR: Must Select Brand before entering VIN



Mfr Dependent VIN select



New Toyota Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for
EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli
list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this
the help file.

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n
indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

4Runner (Toyota) **2003 and later** **Market: 7**

Year	EDR Generation	CDR Cable & Adapter	Module	Module Location
2002 - 2008	Gen 1 (02EDR)	F00K108617 (no adapter)	ACM	Under Center Stack
2009 - 2011	Gen 2 (06EDR)	F00K108616 & F00K108387	ACM	Under Center Stack
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	ACM	Under Center Stack

New Toyota Vehicle Lookup

Camry (Toyota) **2001 and later** **Market: 7**

All Markets the Vehicle is Sold

Year	EDR Generation	CDR Cable & Adapter	M
2001 - 2004 2005 (US/Canada)	Gen 1 (00EDR/02EDR)	F00K108615 (no adapter) F00K108617 (no adapter) Click here [?] to match the ACM part number to the correct cable	A
2005 - 2011	Gen 2 (04EDR/06EDR)	F00K108613 (no adapter) F00K108614 & F00K108387 F00K108615 (no adapter) F00K108616 & F00K108387 Click here [?] to match the ACM part number to the correct cable	A
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	A

< Models

CDR/EDR Training

- Bosch Training Provider Page lists IPTM (Rick Ruth+), Crash Data Specialists (Brad Muir+), CSI (Rusty Haight), CDR-training.com (CSI tech 1&2 trainers) and Northwestern (thru CSI).
- SAE (Rick R) not listed, teaches analysis only.
- There is no “certification” – your qualifications depend on your training and experience
- Bosch does not “certify” any user or endorse any training organization as “preferred”.
- IPTM has ONLINE technician training $295
- All trainers cover basics, but each has unique style and additional content -try them all! END